



BIM PORTFOLIO

G.GOWTHAM, B.E. CIVIL.

BIM ENGINEER

PROJECT SUMMARY – JAKARTA MRT STATION (SENAYAN TO PANCORAN)

ROLE & RESPONSIBILITIES

- Worked as a BIM Engineer for the Jakarta MRT Project, focusing on Station 2 and the Viaduct (Zone J-02).
- Handled parametric modeling, coordination, and documentation using Revit and Tekla.
- Collaborated with a cross-functional team to integrate structural, architectural, and MEP models into a federated BIM environment.

KEY OUTCOMES & ACHIEVEMENTS

- Successfully developed a full coordinated 3D model (LOD 350) for a metro station and viaduct zone.
- Created custom Revit families, reinforcement detailing, and high-quality documentation sheets.
- Conducted clash detection and resolved major inter-disciplinary issues, improving project accuracy.
- Performed 4D construction simulation and 5D quantity tracking for time-cost planning.

TOOLS & SOFTWARE USED

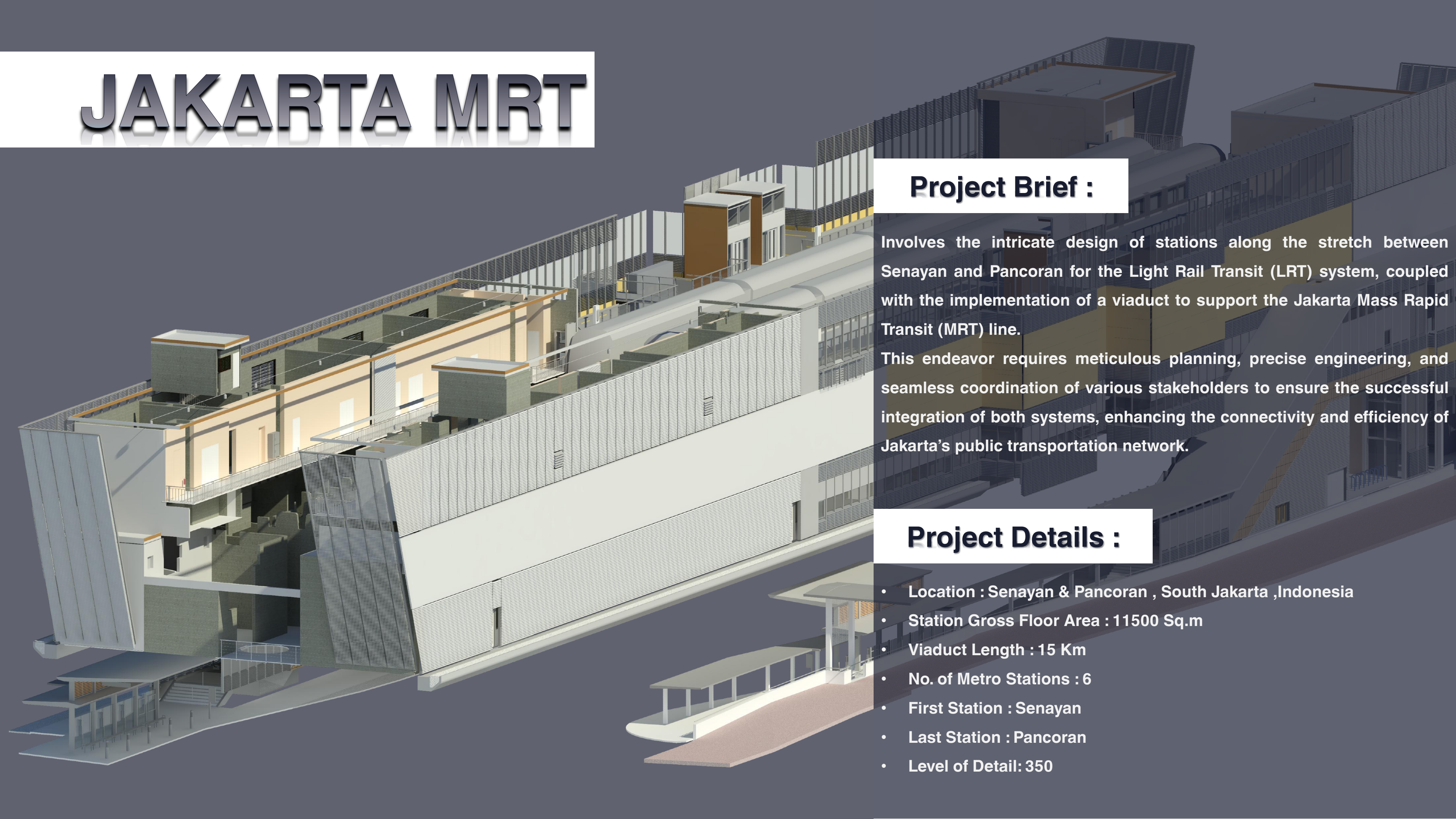
- Revit (Architecture, Structure, Reinforcement, Schedules)
- Civil 3D (Surface generation, Viaduct alignment)
- Navisworks (4D simulation, clash detection, quantity takeoff)
- Tekla Structures (Base plate connections, fabrication drawings)
- Trimble Connect (Model collaboration, CDE workflows)

KEY LEARNINGS

- Applied ISO 19650 workflow for structured BIM execution and delivery.
- Enhanced collaboration through Common Data Environment (CDE) and team-based coordination.
- Gained hands-on knowledge in real-time BIM workflows from design to scheduling.
- Developed strong proficiency in automation, visualization, and construction sequencing.



JAKARTA MRT

A detailed 3D architectural rendering of the Jakarta MRT project. The image shows a long, multi-level viaduct structure with a modern, industrial design. The structure features a series of rectangular sections connected by walkways and stairs. The color palette is primarily grey and white, with some orange and blue accents. The rendering is set against a dark blue background, and the perspective is from a low angle, looking up at the structure.

Project Brief :

Involves the intricate design of stations along the stretch between Senayan and Pancoran for the Light Rail Transit (LRT) system, coupled with the implementation of a viaduct to support the Jakarta Mass Rapid Transit (MRT) line.

This endeavor requires meticulous planning, precise engineering, and seamless coordination of various stakeholders to ensure the successful integration of both systems, enhancing the connectivity and efficiency of Jakarta's public transportation network.

Project Details :

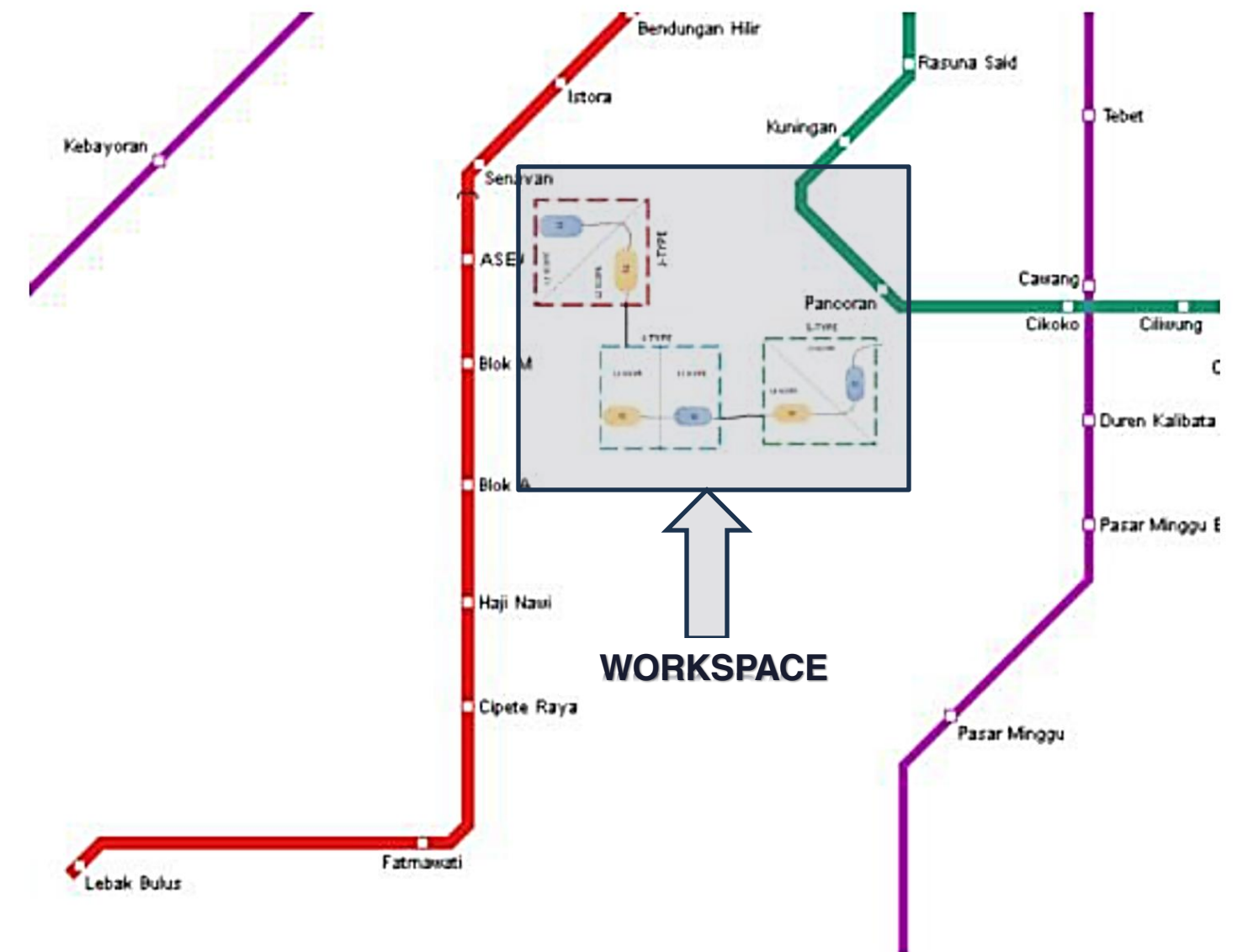
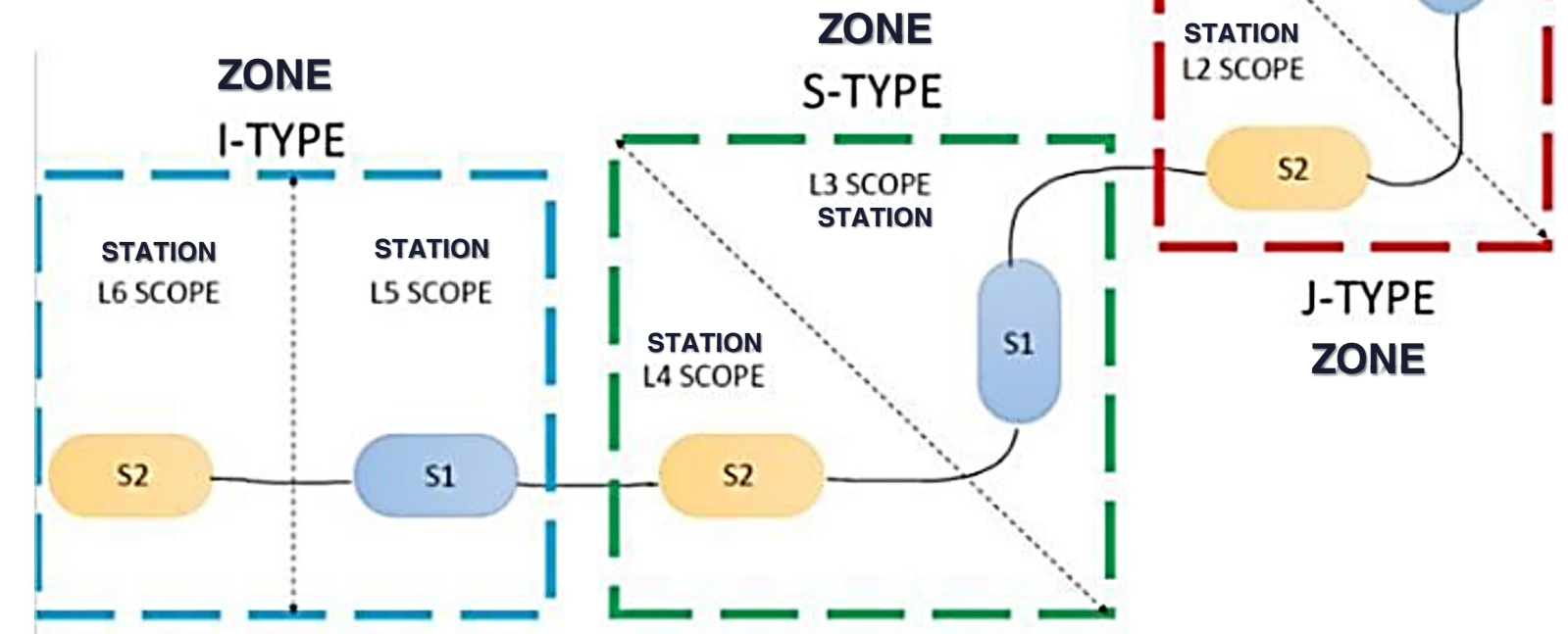
- Location : Senayan & Pancoran , South Jakarta ,Indonesia
- Station Gross Floor Area : 11500 Sq.m
- Viaduct Length : 15 Km
- No. of Metro Stations : 6
- First Station : Senayan
- Last Station : Pancoran
- Level of Detail: 350

TEAM & ZONE

- The workspace is divided into 6 stations (L-1 to L-6).
- These stations are grouped into 3 zones (TYPE - J, S & I).
- Each zone is managed by 2 team members.
- This team-zone structure helps streamline workflow.
- It improves communication among team members.
- It fosters a professional and productive environment.

S.No	BIM ENGINEER	STATION NO	ZONE ALLOTTED
1	PRATIK	L - 1	J-Type
2	GOWTHAM	L - 2	J-Type
3	VIVEK	L - 3	S-Type
4	LAIQUE RAZA	L - 4	S-Type
5	VEDAVATHI	L - 5	I-Type
6	SACHIN	L - 6	I-Type

WORKSPACE



ISO 19650 STAGES AND KEY LEARNINGS

APPOINTMENT 04

In this stage you will create construction drawings and technical details

MOBILIZATION 05

In this stage you will handover your construction drawings to on-site

COLLABORATIVE PRODUCTION OF INFORMATION 06

This stage you will understand the Handover process

INFORMATION MODEL DELIVERY 07

In this stage you will see the building in use

PROJECT CLOSE-OUT 08

In this stage you will get your portfolio reviewed by career guide

00 CLIENT BRIEF

Establishing project understanding and identifying stakeholders.

01 ASSESSMENT AND NEED

To assemble the project team; define the structure, team roles, and responsibilities.

02 INVITATION TO TENDER

Appointing party creates tender docs for project bidding.

03 TENDER RESPONSE

In this Stage you will work on Tender response

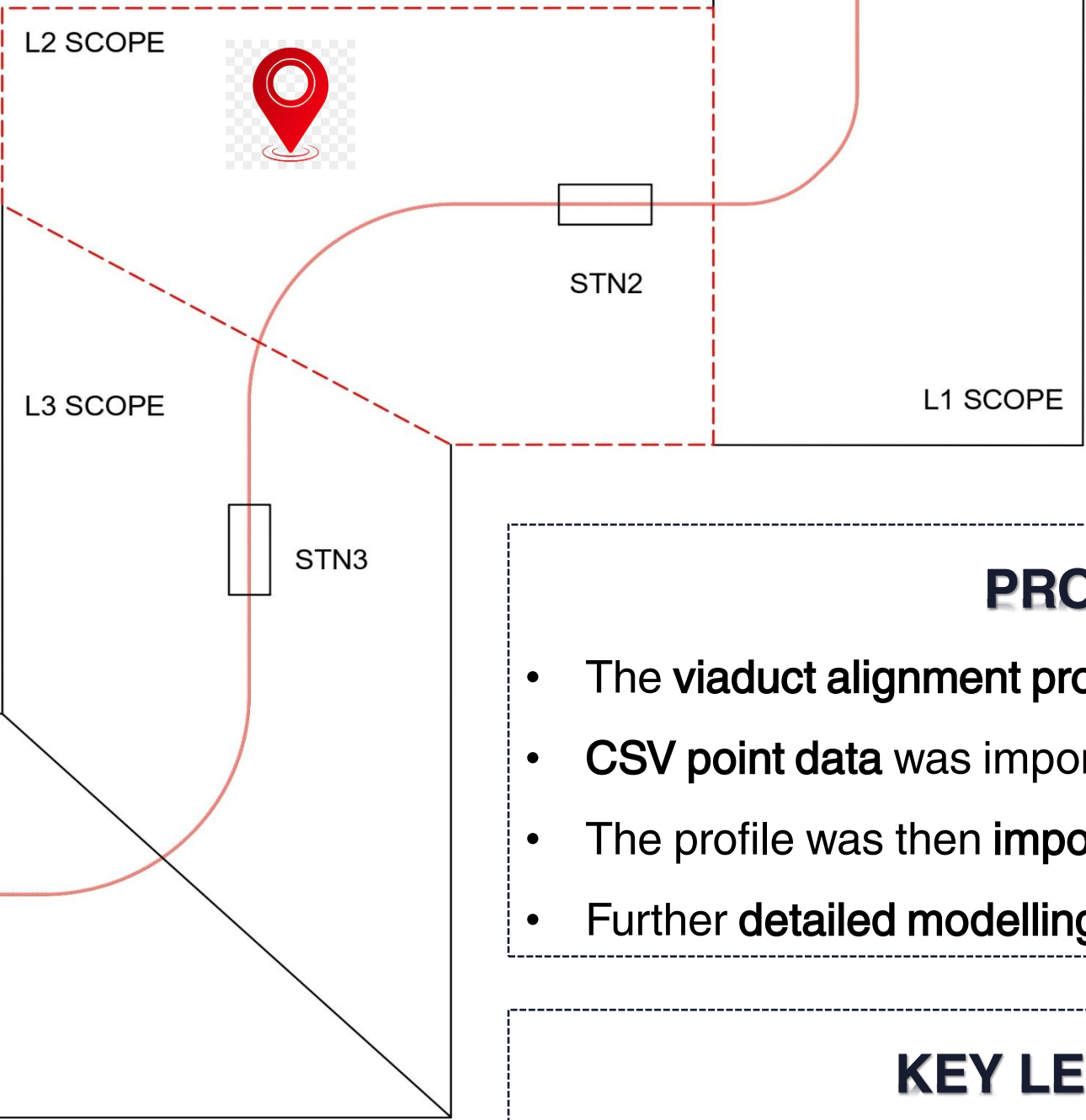


VIADUCT ALIGNMENT

ZONE AND STATION ASSIGNMENT DETAILS

- This is part of a **large MRT project** connecting **Senayan Metro** and **Pancoran LRT** stations.
- The project includes construction of **6 new metro stations**.
- I will be focusing on **STN 2**, located in the **red-marked zone**.
- The **rest of the team** will handle the **remaining stations**.
- This team approach ensures a **seamless workflow** and **high-quality results** across the entire project.

MY ZONE & STATION



CONNECTING TO
PANCORAN LRT

SOFTWARE USED



Civil 3D

PROCESS

- The **viaduct alignment profile** was created using **Civil 3D**.
- **CSV point data** was imported to generate the **surface**.
- The profile was then **imported into Revit**.
- Further **detailed modelling** was carried out in **Revit**.

KEY LEARNINGS

- Imported **survey data** and created **surface** in **Civil 3D**.
- Developed **alignment** for design layout.
- Enhanced **data handling** and **workflow accuracy**.

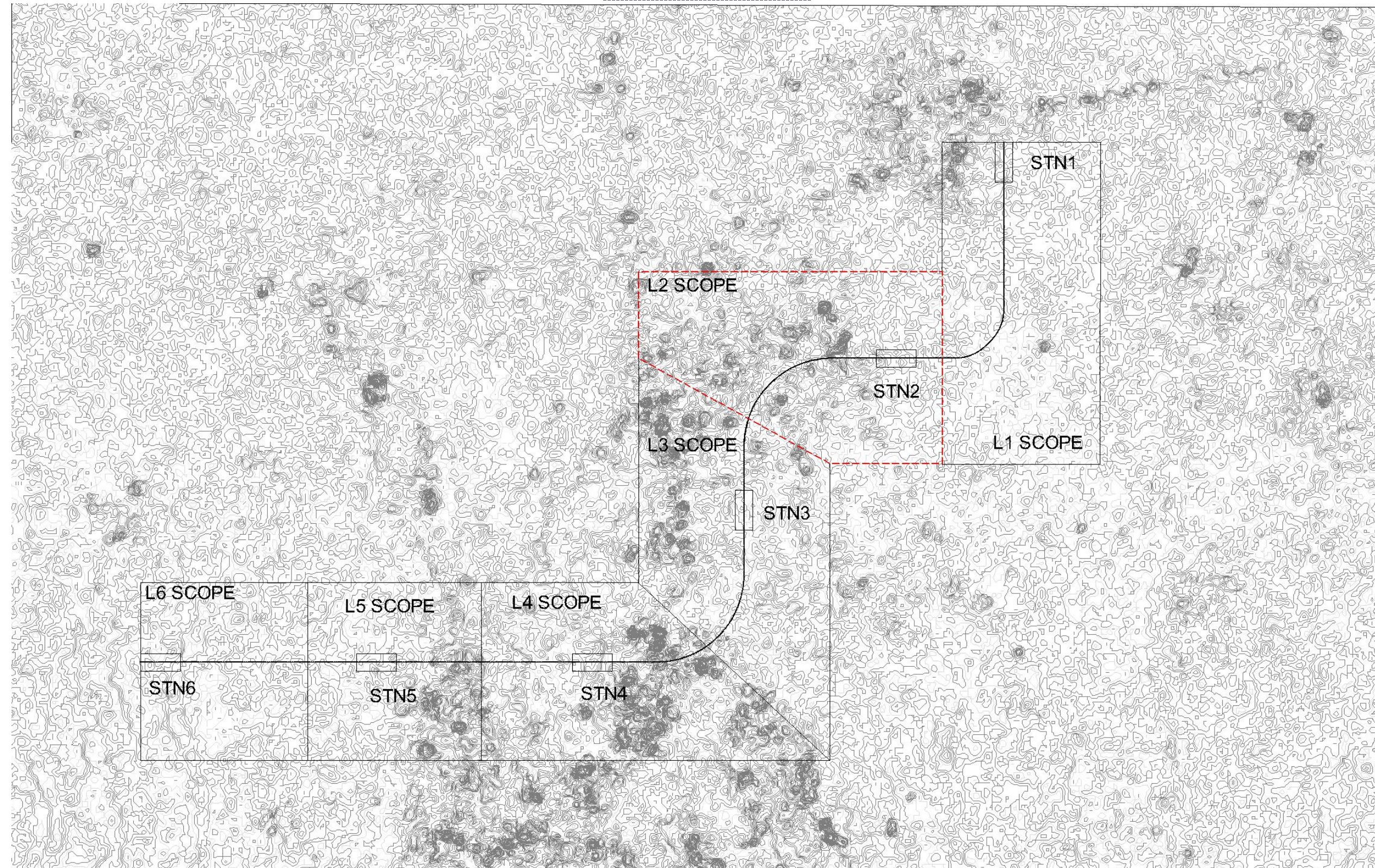
CONNECTING TO
SENAYAN METRO

CIVIL 3D - VIADUCT ALIGNMENT

OUTPUT

PROCESS

- Ground points were imported into Civil 3D using a CSV file.
- A surface was generated based on the imported points.
- The best possible alignment was fixed after studying the profile.

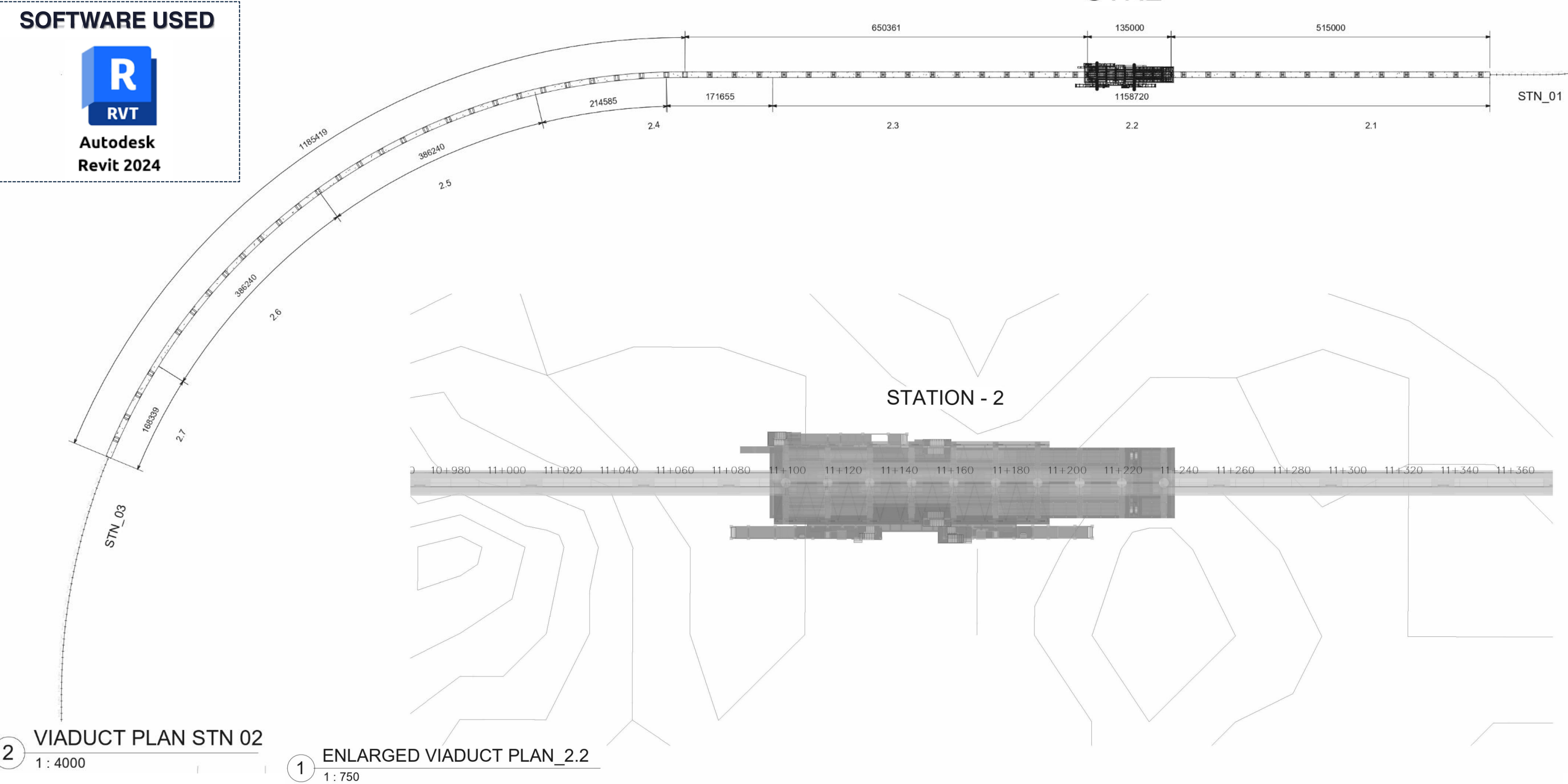


STATION 02 - VIADUCT DETAILS

SOFTWARE USED



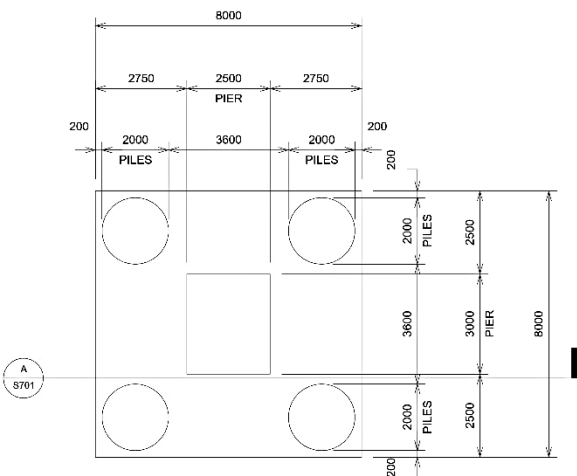
Autodesk
Revit 2024



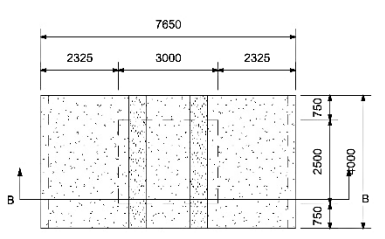
2 VIADUCT PLAN STN 02
1 : 4000

1 ENLARGED VIADUCT PLAN_2.2
1 : 750

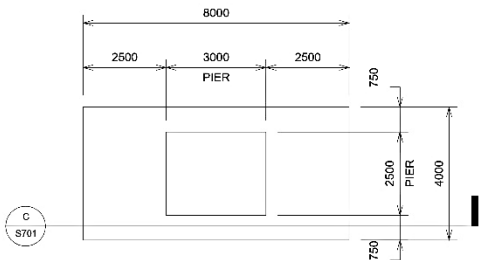
VIADUCT – PILE, PIER & PIER CAP



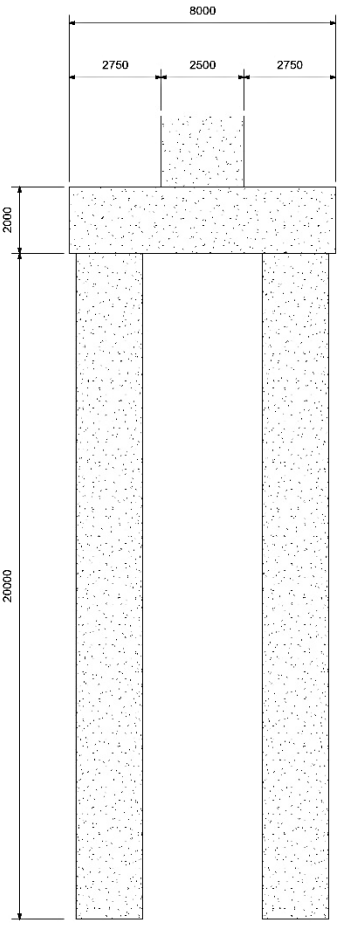
1 PILE CAP
1 : 100



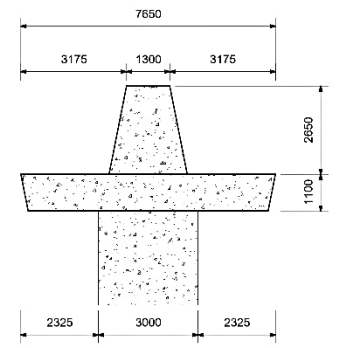
2 PIER CAP-1
1 : 100



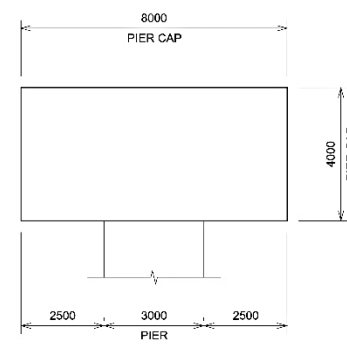
3 PIER CAP-2
1 : 100



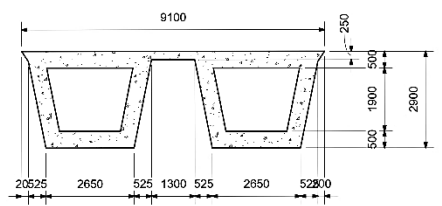
A PILE CAP SECTION
1 : 100



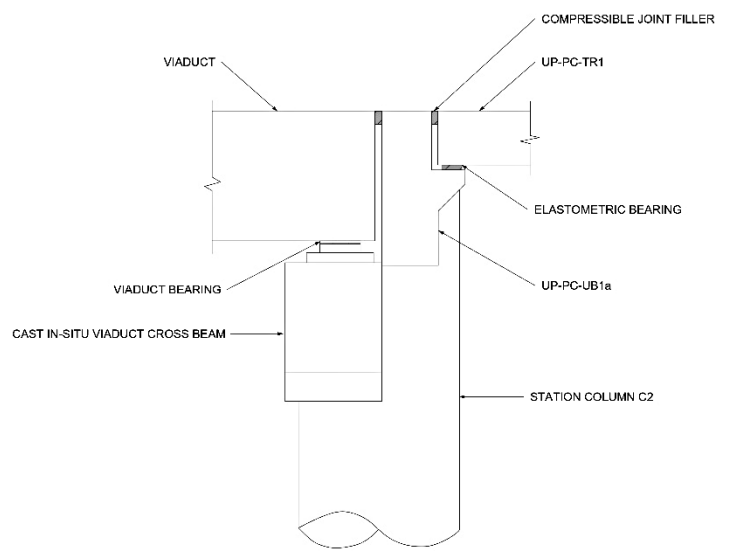
B PIER CAP-1 SECTION
1 : 100



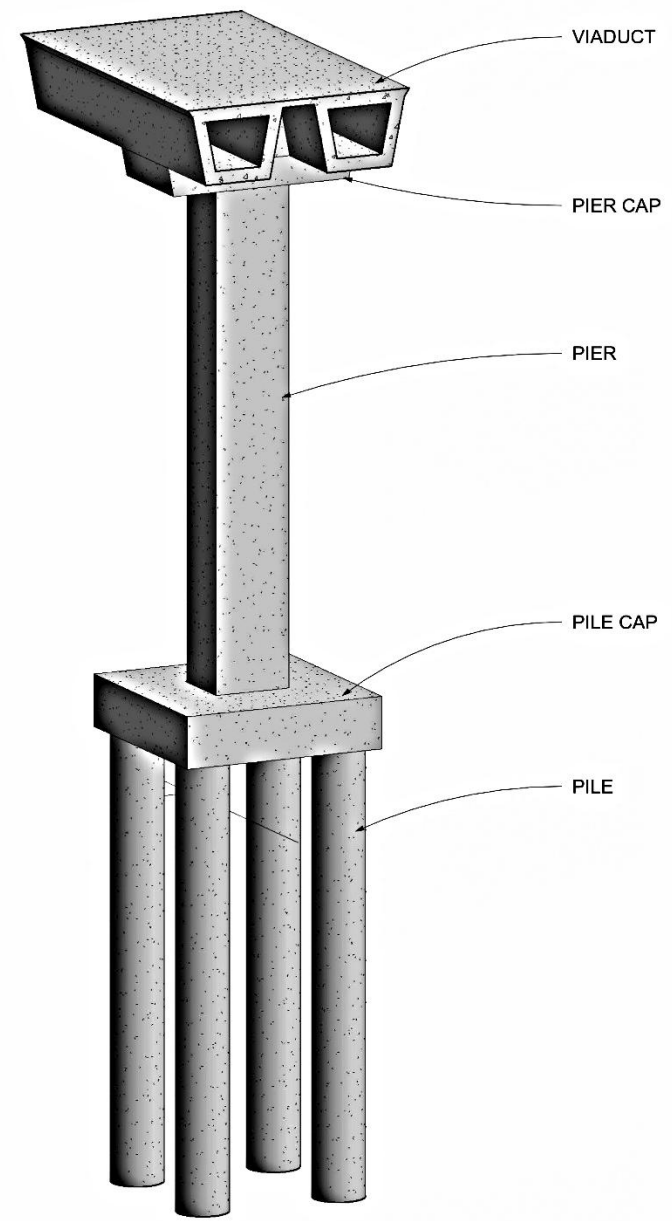
C PIER CAP-2 SECTION
1 : 100



4 VIADUCT CROSS SECTION
1 : 100



5 VIADUCT STATION CONNECTION INTERFACE
1 : 100



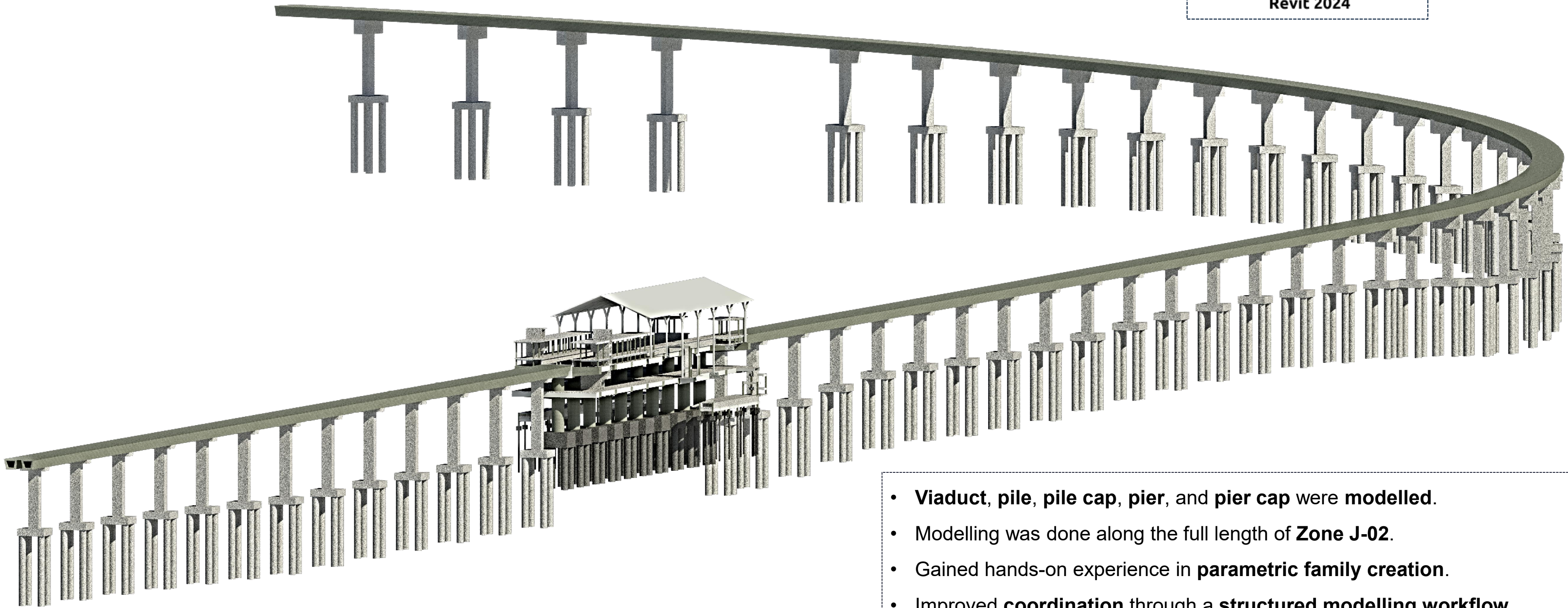
1 ISOMETRIC SECTION

VIADUCT – RENDERED VIEW

SOFTWARE USED

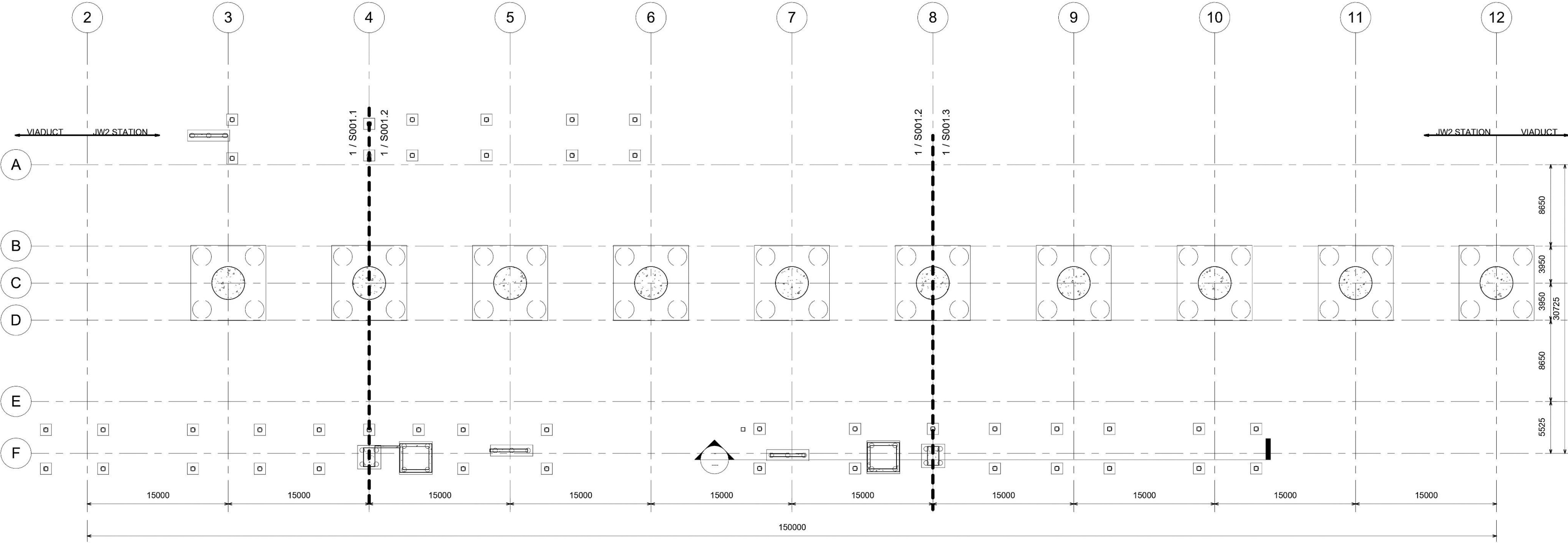


Autodesk
Revit 2024

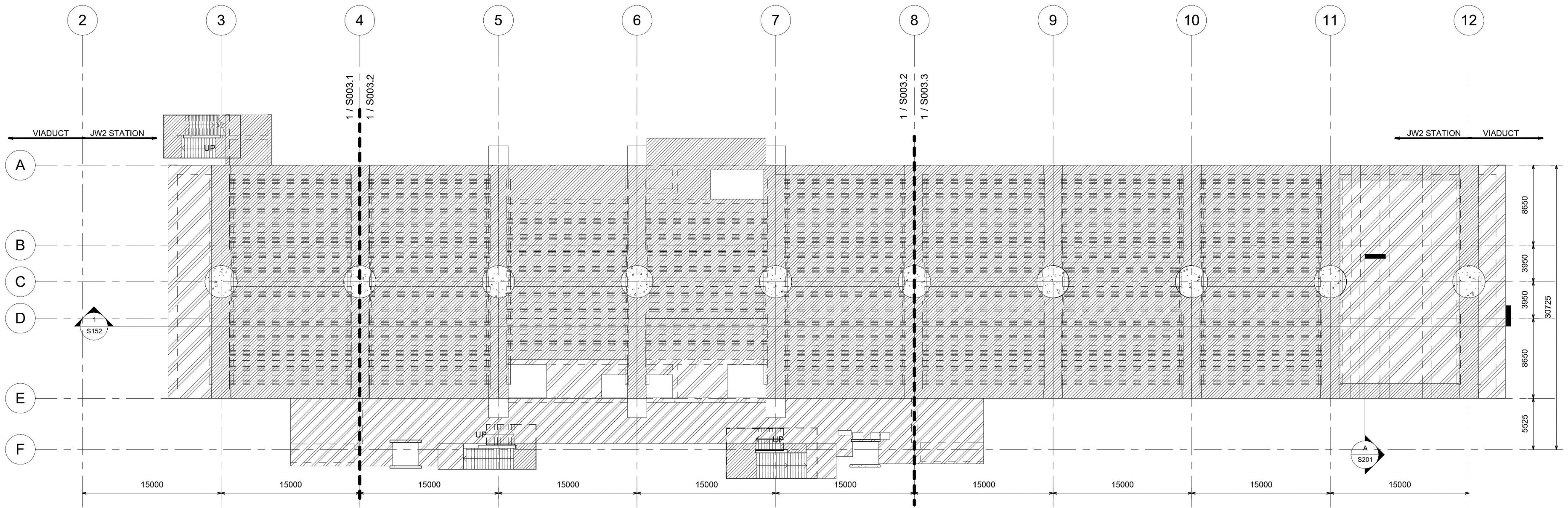


- **Viaduct, pile, pile cap, pier, and pier cap** were modelled.
- Modelling was done along the full length of **Zone J-02**.
- Gained hands-on experience in **parametric family creation**.
- Improved **coordination** through a **structured modelling workflow**.
- Learned the importance of **accuracy** and **Level of Detail (LOD)** in **BIM**.
- Understood how to handle **complex structures** like **viaducts**.

TOP OF PILE CAP PLAN



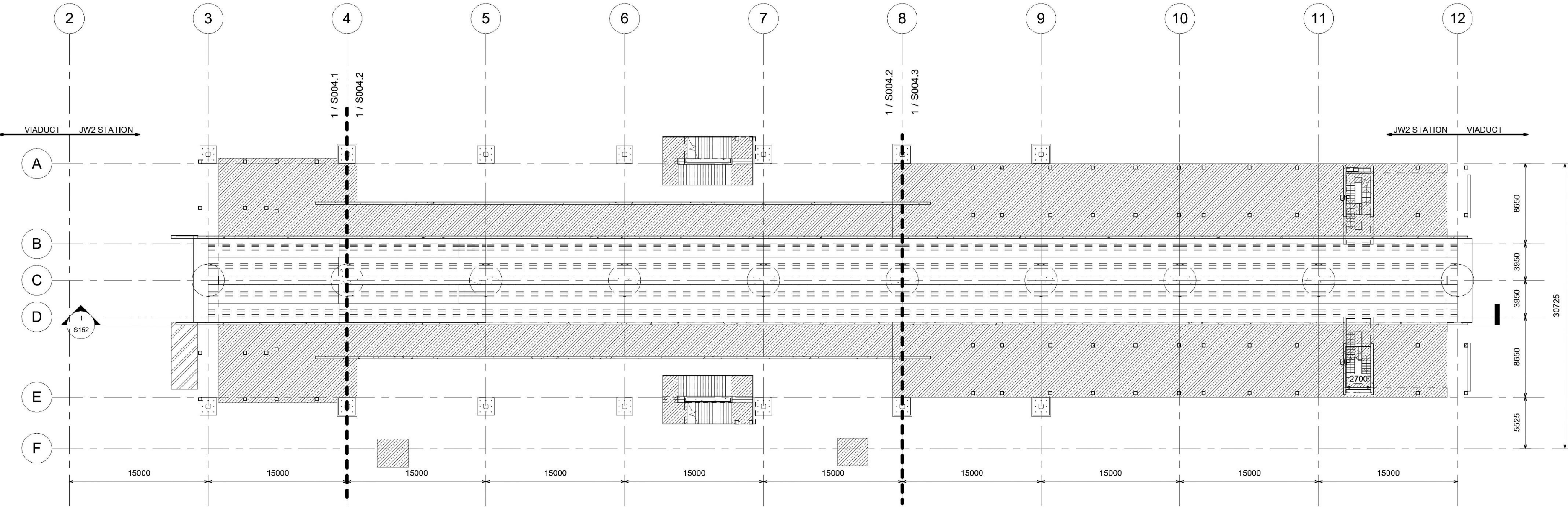
CONCOURSE LEVEL PLAN



LEGENDS

-  SLAB 100MM THICK
-  SLAB 250MM THICK
-  SLAB 300MM THICK

UNDER PLATFORM LEVEL PLAN

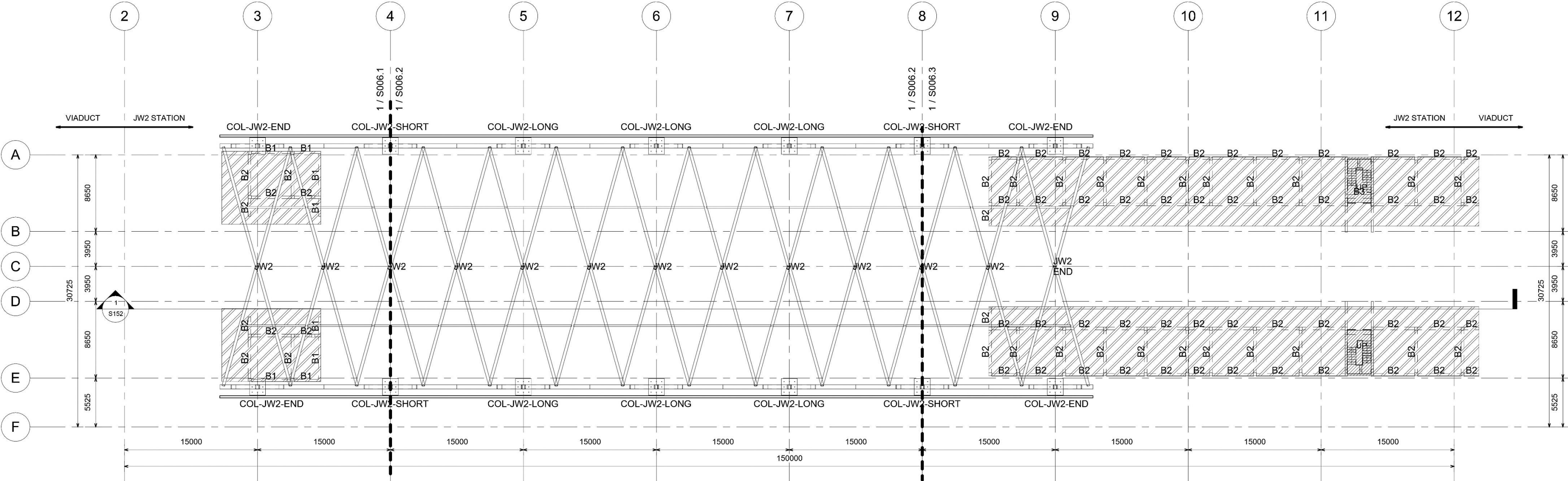


LEGENDS

SLAB 100MM THICK

SLAB 250MM THICK

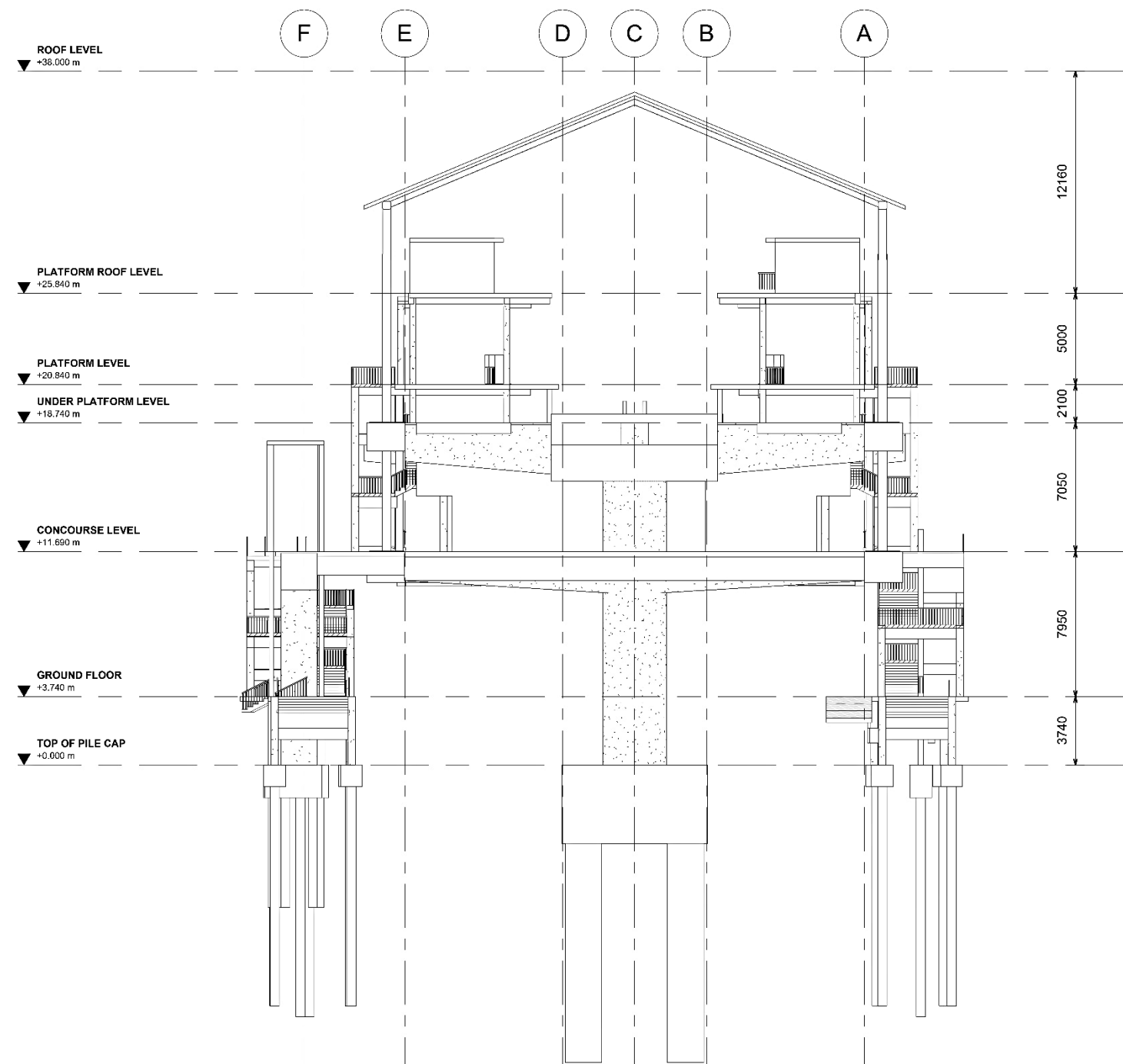
ROOF PLAN



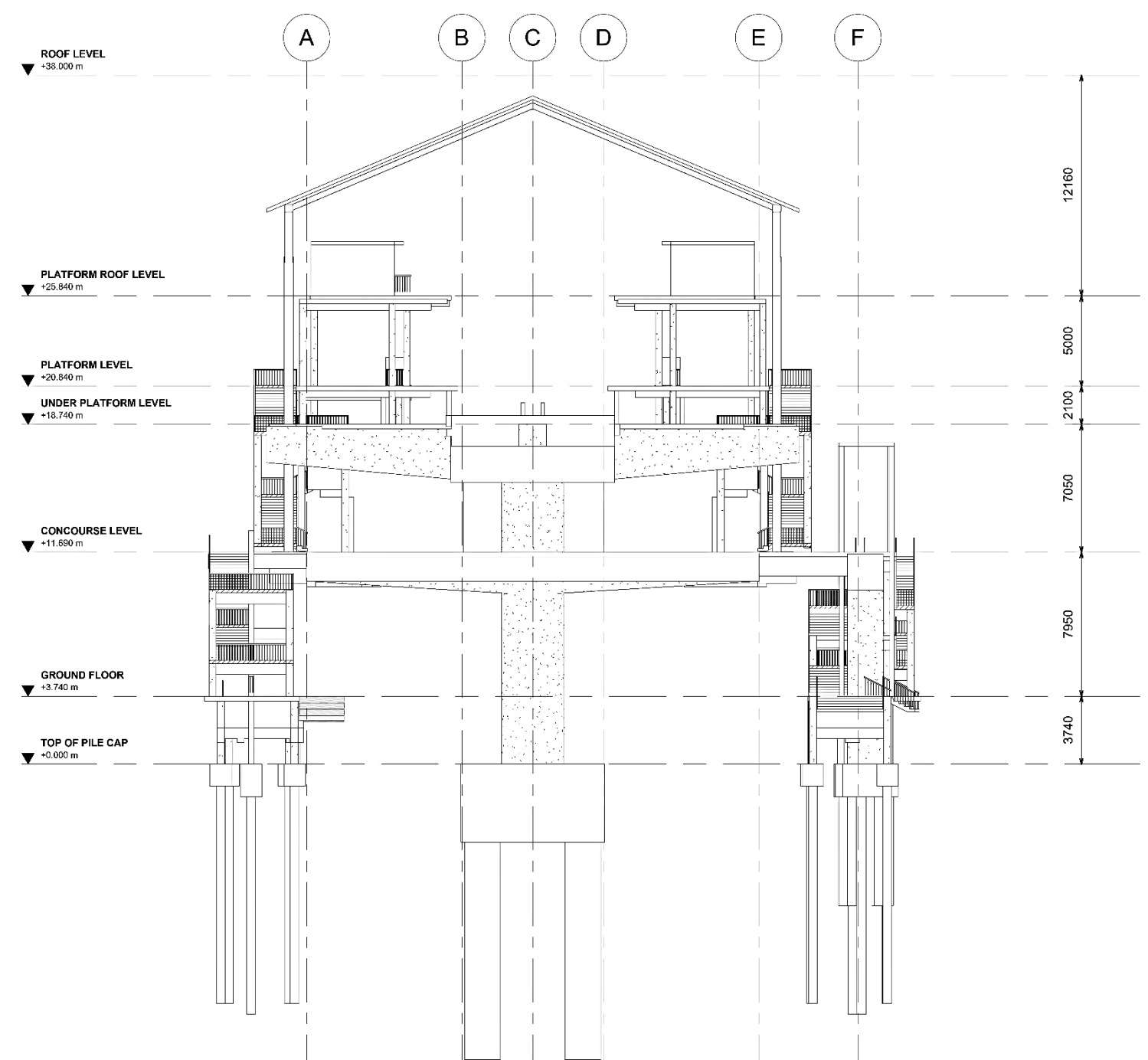
LEGENDS

 SLAB 100MM THICK SLAB 250MM THICK

ELEVATIONS



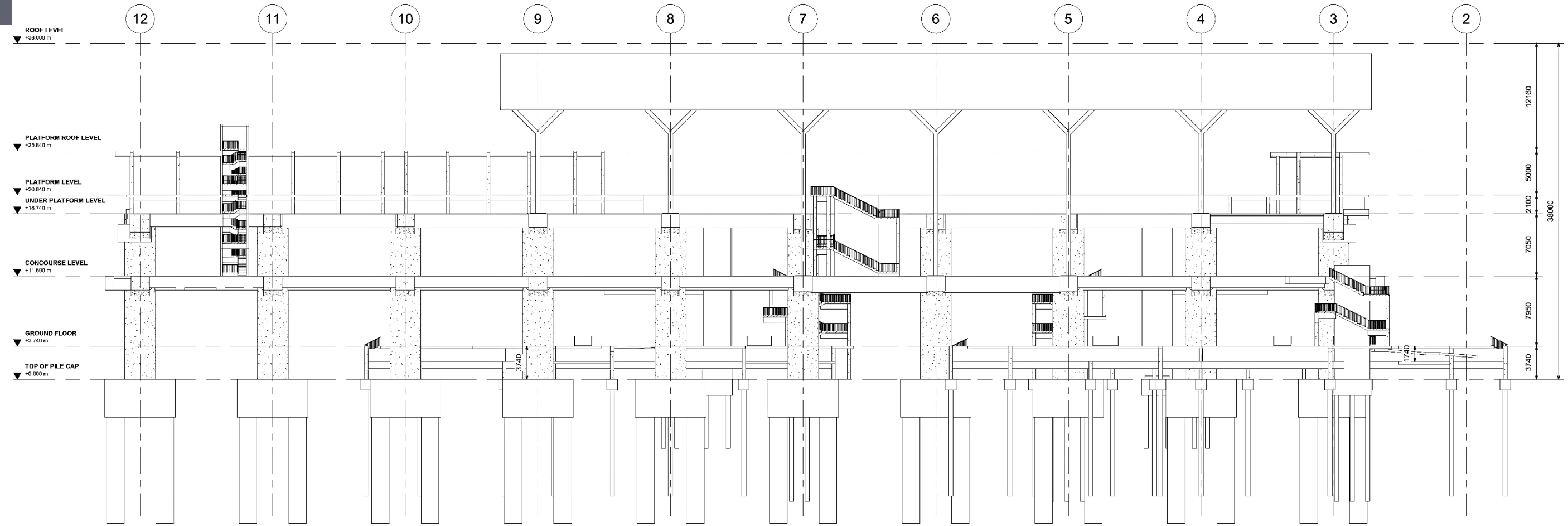
EAST



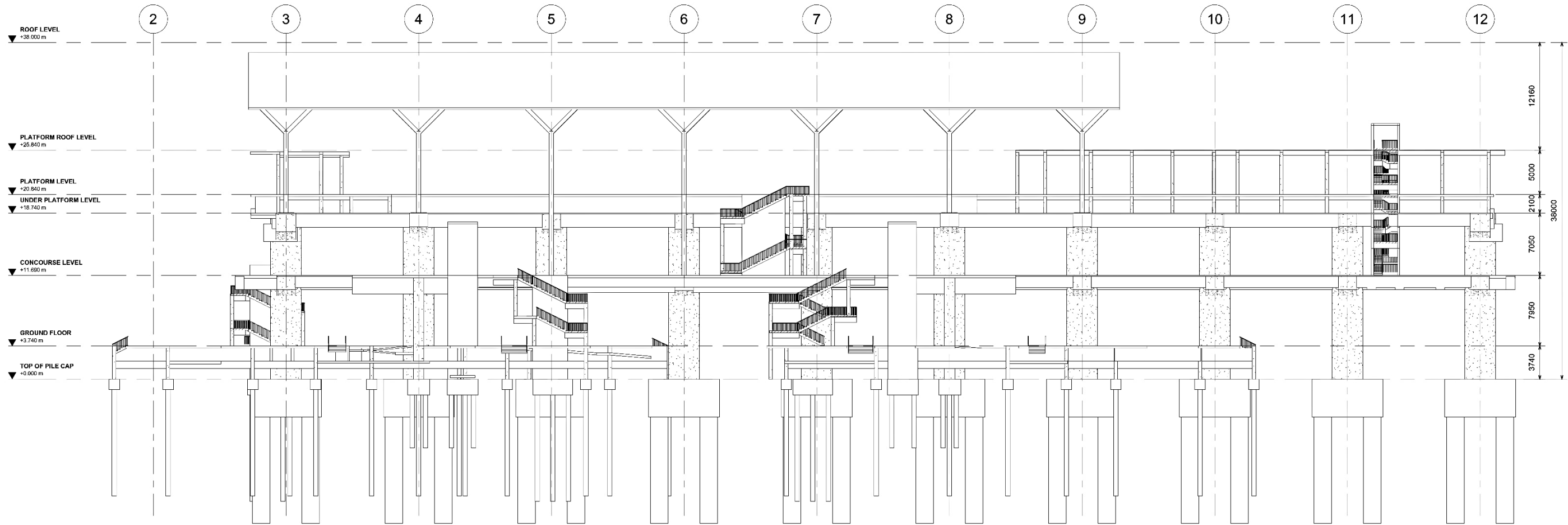
WEST

ELEVATIONS

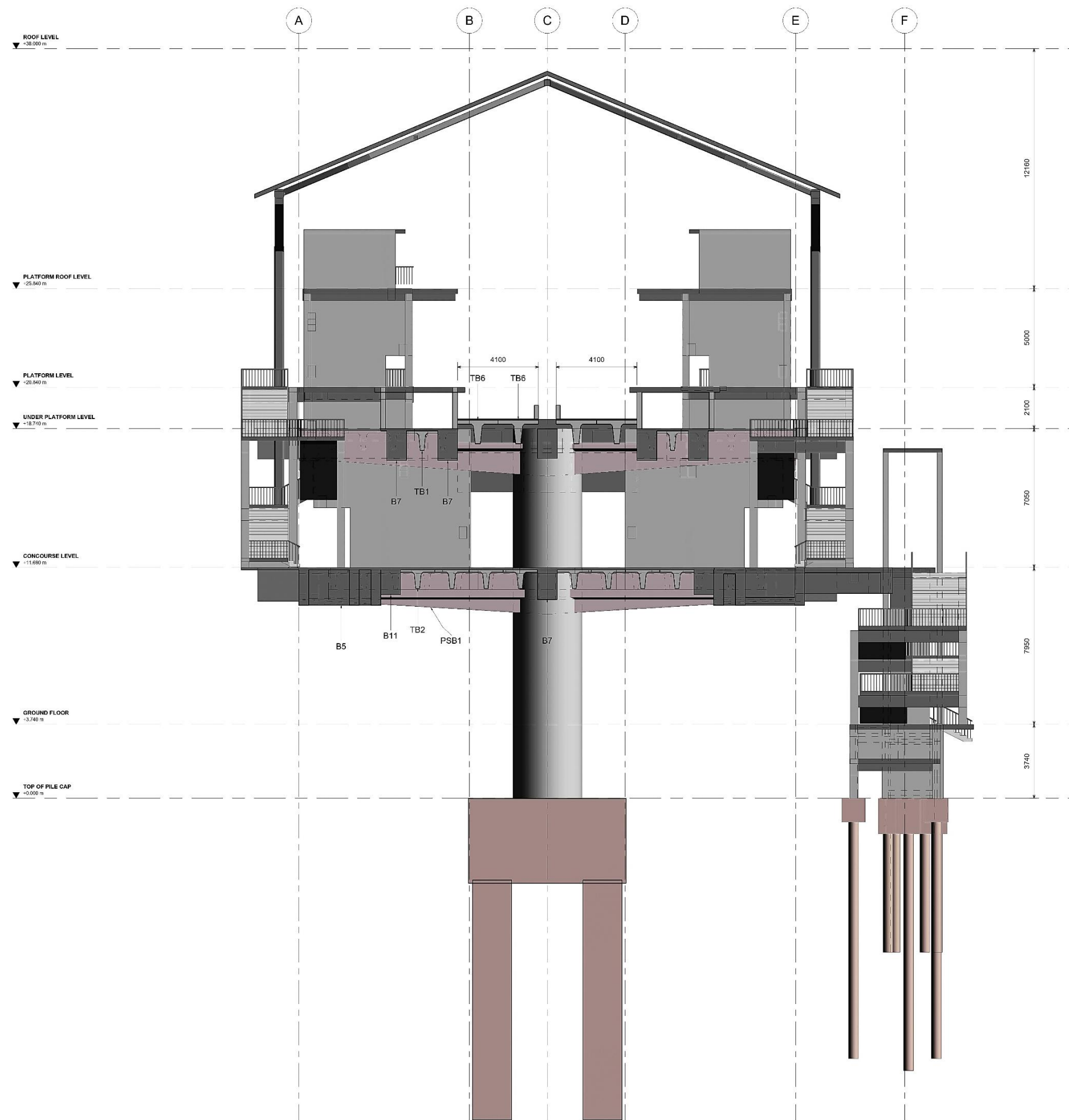
NORTH



SOUTH

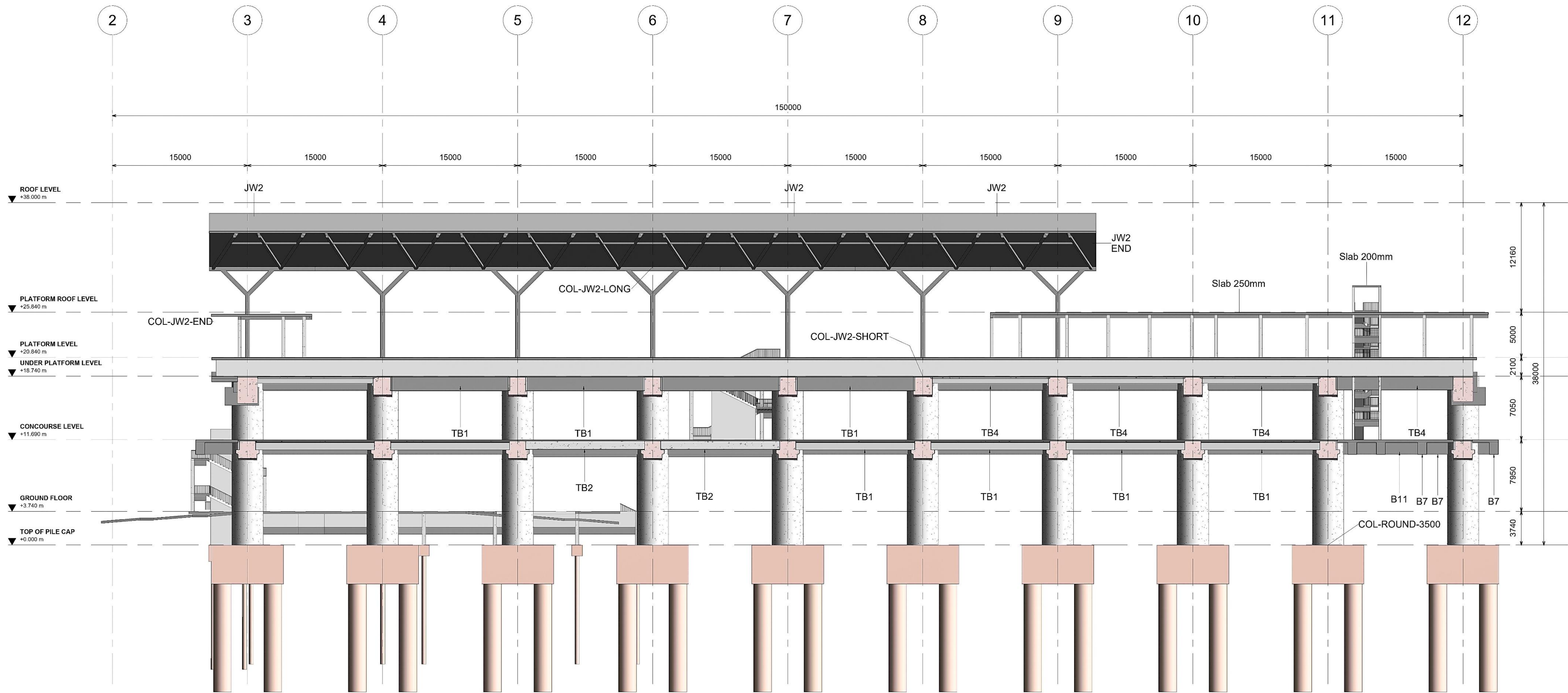


TRANSVERSE SECTION

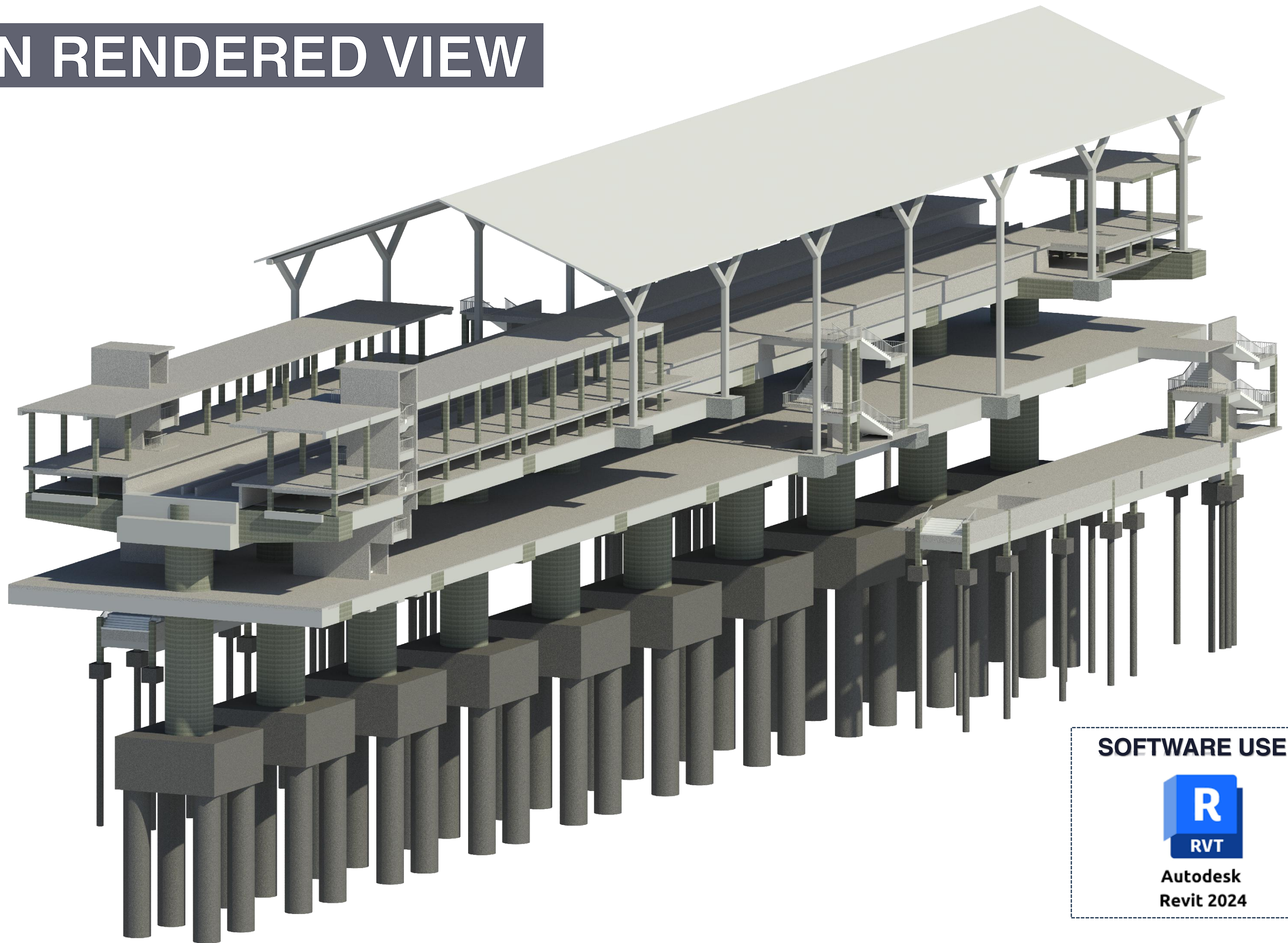


- **Section Tool** in Revit allows for quick generation of **detailed sectional views** at any required location.
- Helps to **analyze internal components**, such as structural systems, platform levels, and foundation elements with clarity.
- Enhances **coordination** between disciplines by visualizing the **complete cross-section** for better decision-making.

LONGITUDINAL SECTION



STATION RENDERED VIEW

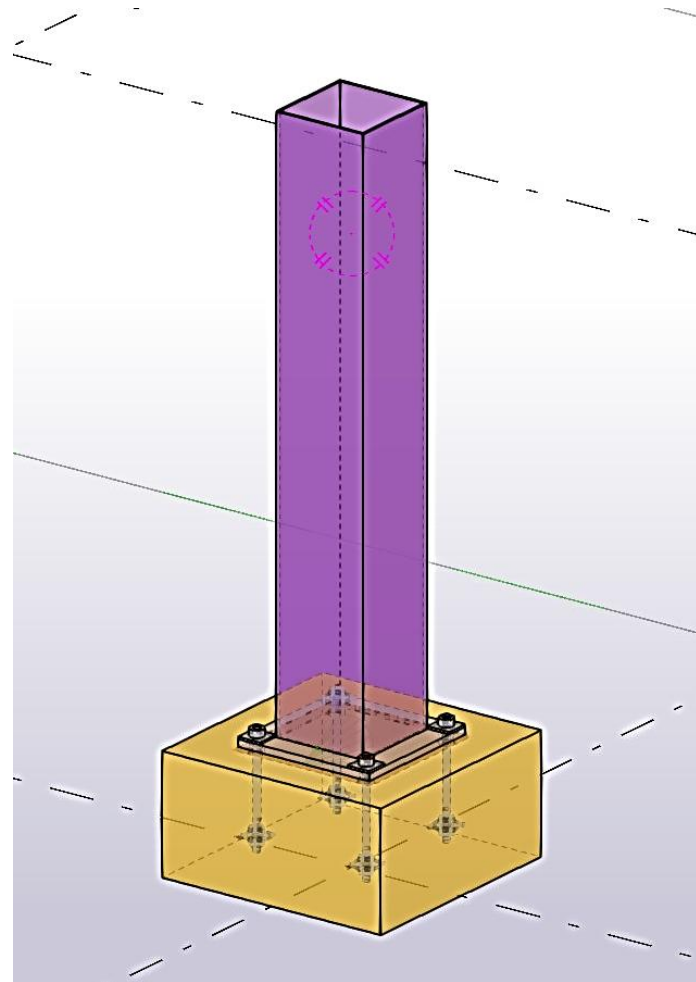


SOFTWARE USED

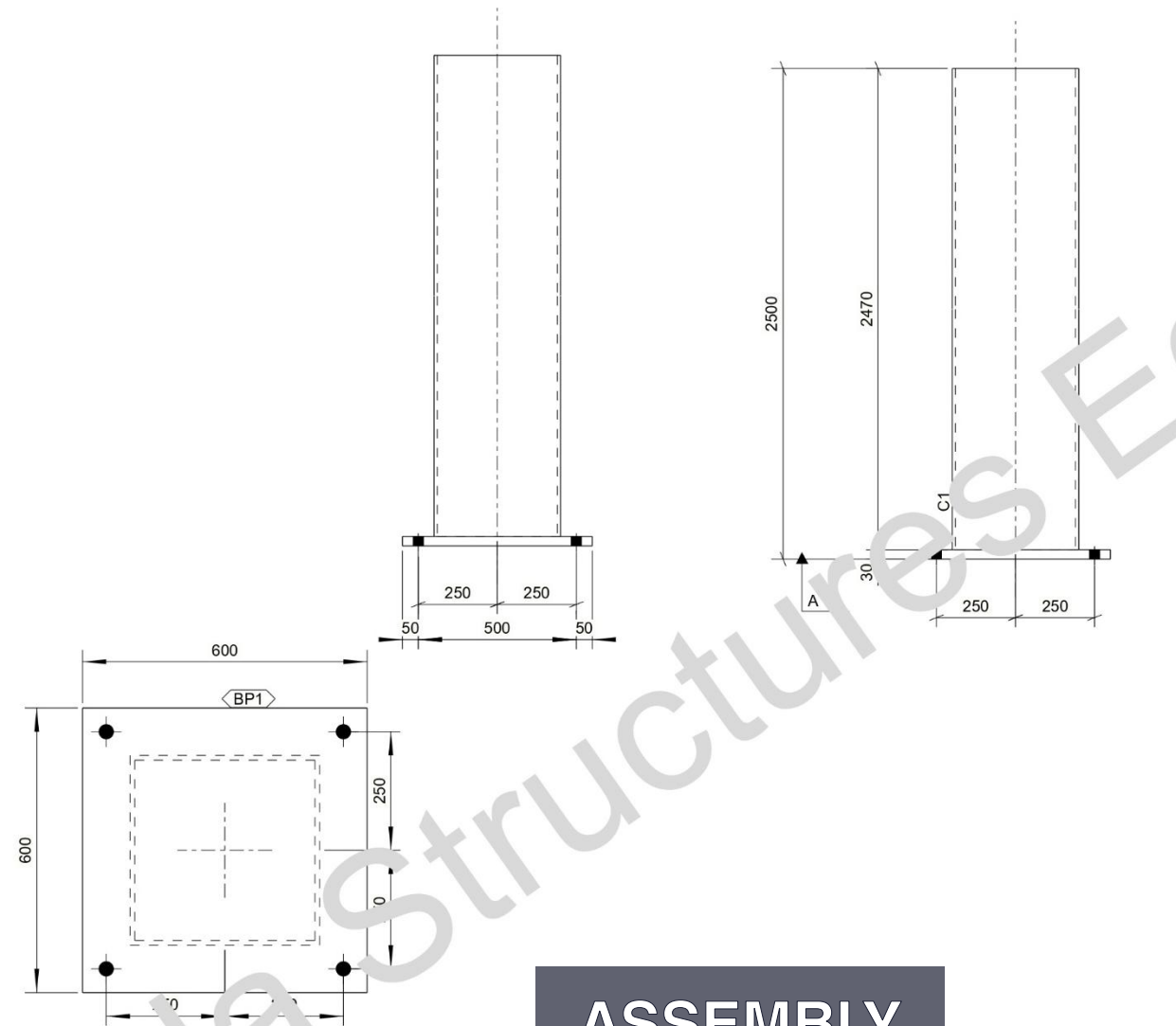


Autodesk
Revit 2024

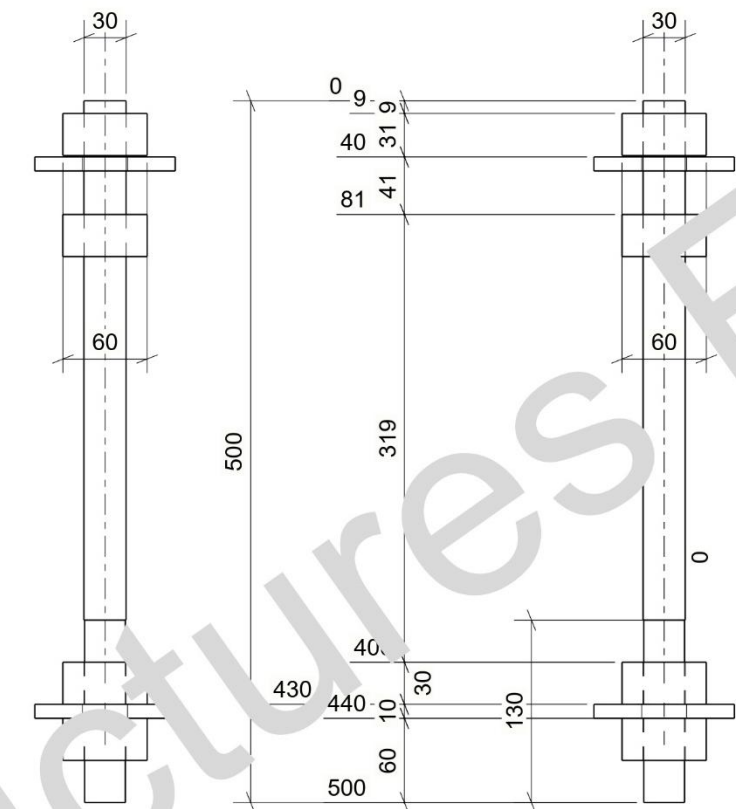
MODEL STEEL BASE PLATE CONNECTION BETWEEN THE STEEL COLUMN & CONCRETE COLUMN



3D VIEW



ASSEMBLY
DRAWING



SINGLE PART
DRAWING

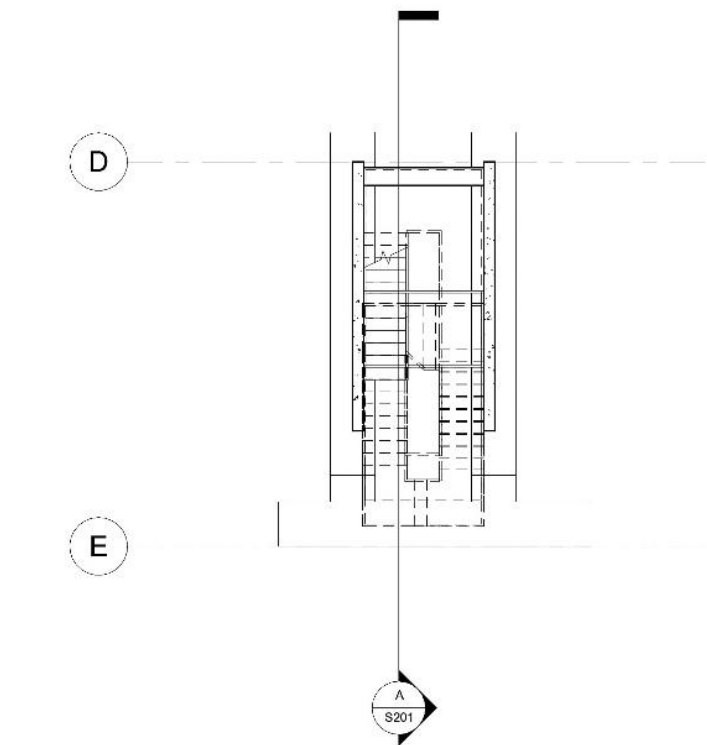


- Model steel base plate connection between steel and concrete column in Tekla Structures.
- Visualize the connection using 3D view.
- Generate Assembly Drawings for full fabrication.
- Create Single Part Drawings for individual components.
- Apply accurate dimensions and bolt spacing.
- Use Tekla for detailed modeling and constructible outputs.

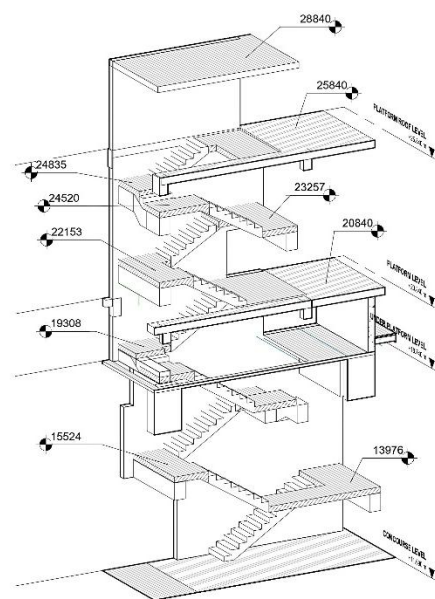
SOFTWARE USED



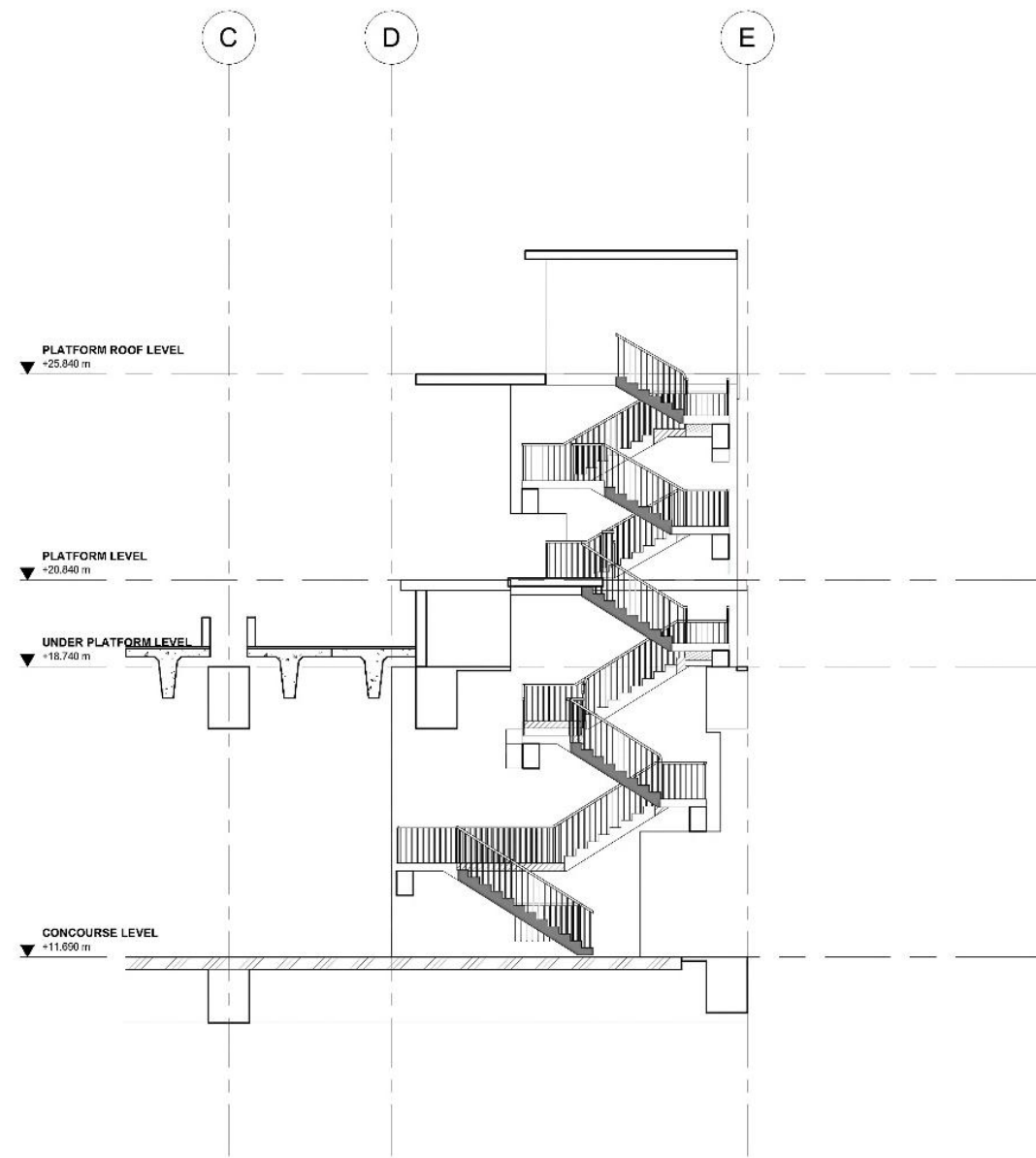
SHEET DOCUMENTATION - STAIRCASE



2 CONCOURSE LEVEL - STAIRCASE
1 : 100



1 Staircase 3D



A Section
1 : 100

GENERAL NOTES
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2. All drawings are to be read in conjunction with other relevant drawings and specifications of the same and other disciplines.
3. This drawing is not to be scaled. Only written dimensions are to be followed. Any discrepancy must be advised to the Architect and the Engineer.
4. All dimensions are metric unless otherwise indicated.
5. The walls are dimensioned to the unfinished face of masonry or concrete unless otherwise noted.
6. Final coordination is the responsibility of the contractor. Any discrepancy must be brought to the attention of the Architect and the Engineer and must be resolved before commencement of work.

Rev. Description Issue Date
Lead Design / BM Consultant

NOVATR

Client
Ministry of Public Works and Housing
Jalan Pattimura No. 20 Kebayoran Baru
Jakarta Selatan, 12110
Jakarta, Indonesia

Project Name
**DESIGN AND CONSTRUCTION OF STATIONS
BETWEEN SENAYAN & PANCORAN LRT AND VIADUCT
FOR JAKARTA MRT LINE**

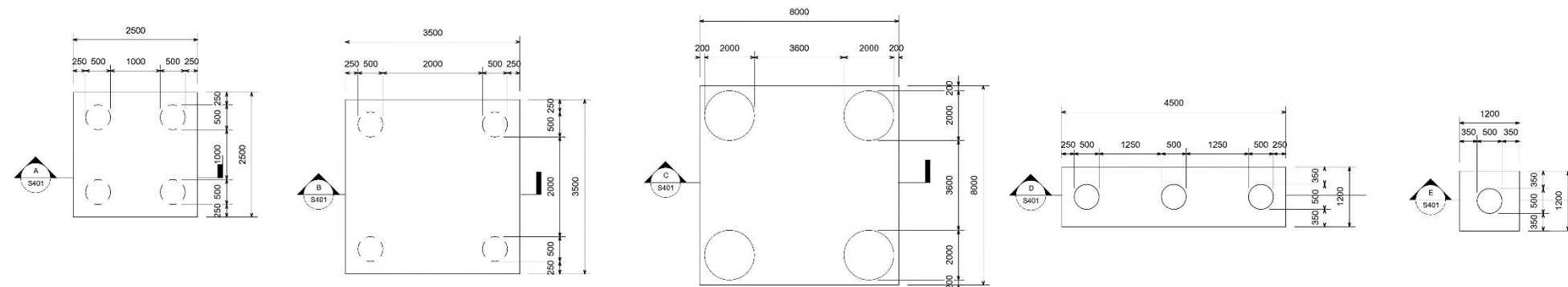
SENAYAN & PANCORAN, SOUTH JAKARTA, INDONESIA

Drawing Title
STAIRCASE

Stage: Project Status Date: 05/31/25 Drawn: Author
Project No: J056 Scale: 1:100 @ A1 Check: Checker
Drawing No: J056 - CAP - STR - S201 Rev:

- Title block creation is done to define the project identity and standardize drawing presentation.
- Sheets are created using the Sheet Tool in Revit by placing the customized title block.
- These sheets are used to organize views like plans, sections, and 3D views for clear documentation.

SHEET DOCUMENTATION – PILE & PILE CAP



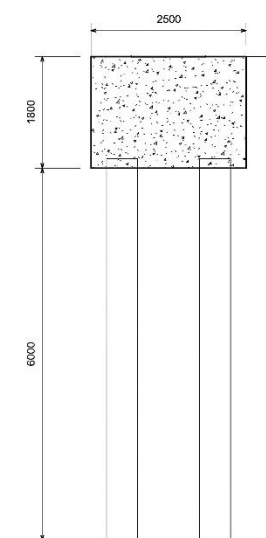
1 P1
1 : 50

2 P2
1 : 50

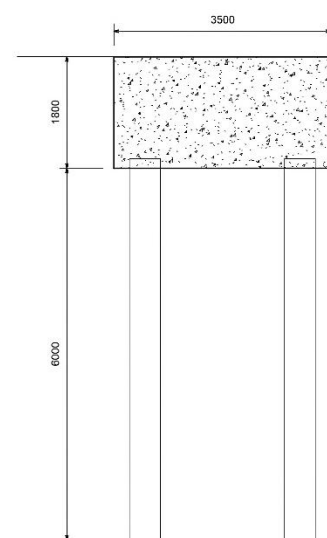
3 P3
1 : 100

4 P4
1 : 50

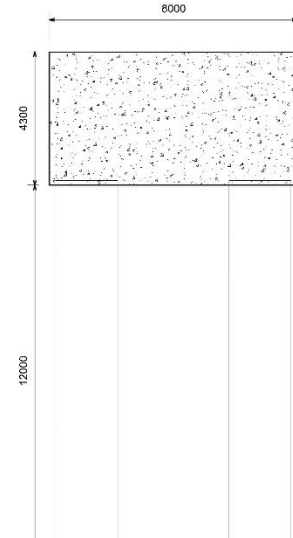
5 P5
1 : 50



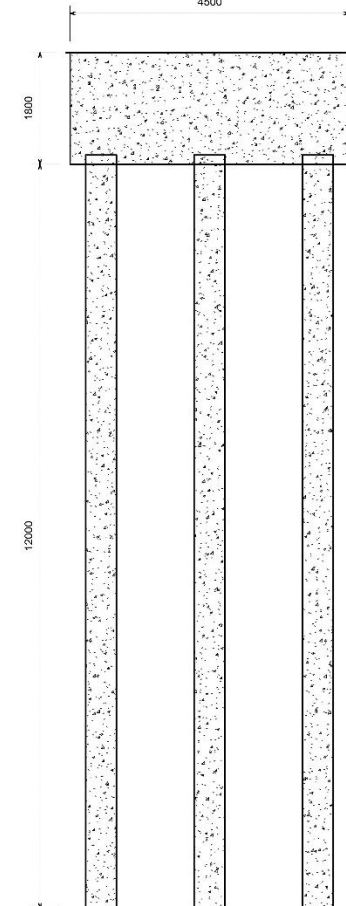
Section
1 : 50



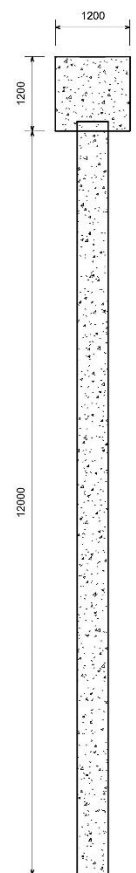
Section
1 : 50



Section
1 : 100



D Section
1 : 50



E Section
1 : 50

Pile Cap Schedule			
Type Mark	Length	Width	Foundation Thickness
P1	2500	2500	1800
P2	3500	3500	1800
P3	8000	8000	4300
P4	4500	1200	1800
P5	1200	1200	1200
Grand total: 60			

GENERAL NOTES

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6. For the determination of the responsibility of the contractor, Any discrepancy must be brought to the attention of the Architect and the Engineer and must be resolved before commencement of work.

- A single parametric family was created for P1, P2, and P3, allowing dimension changes without duplicating models.
- **Type variations** were efficiently managed by adjusting **parameters** like length, width, and depth.
- For P4 and P5, separate families were created due to their distinct **geometry and design requirements**.

Rev	Description	Issue Date
1	Lead Design / BIM Consultant	

NOVATE

Client	
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Project Name
**DESIGN AND CONSTRUCTION OF STATIONS
BETWEEN SENAYAN & PANCORAN LRT AND VIADUCT
FOR JAKARTA MRT LINE**

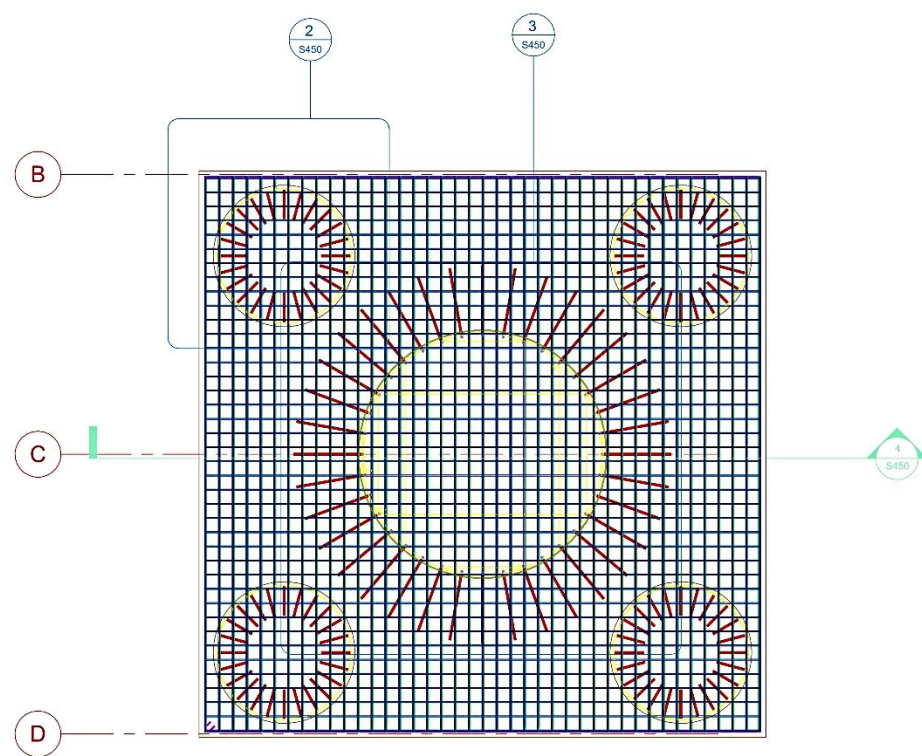
SENAYAN & PANCORAN, SOUTH JAKARTA, INDONESIA

Drawing Title

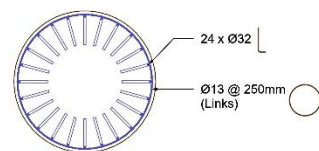
DETAILS PILE FOUNDATION FAMILY

Stage: Project Status	Date: 06/01/25	Drawn: Author
Project No: J056	Scale: As indicated	Check: Checker
Drawing No: J056 - CAP - STR - S401		Rev.

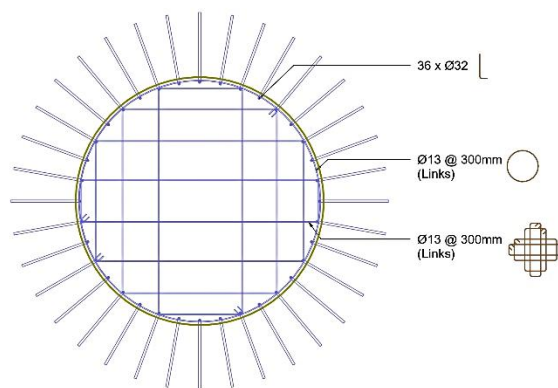
REINFORCEMENT DETAILING – PILE & PILE CAP



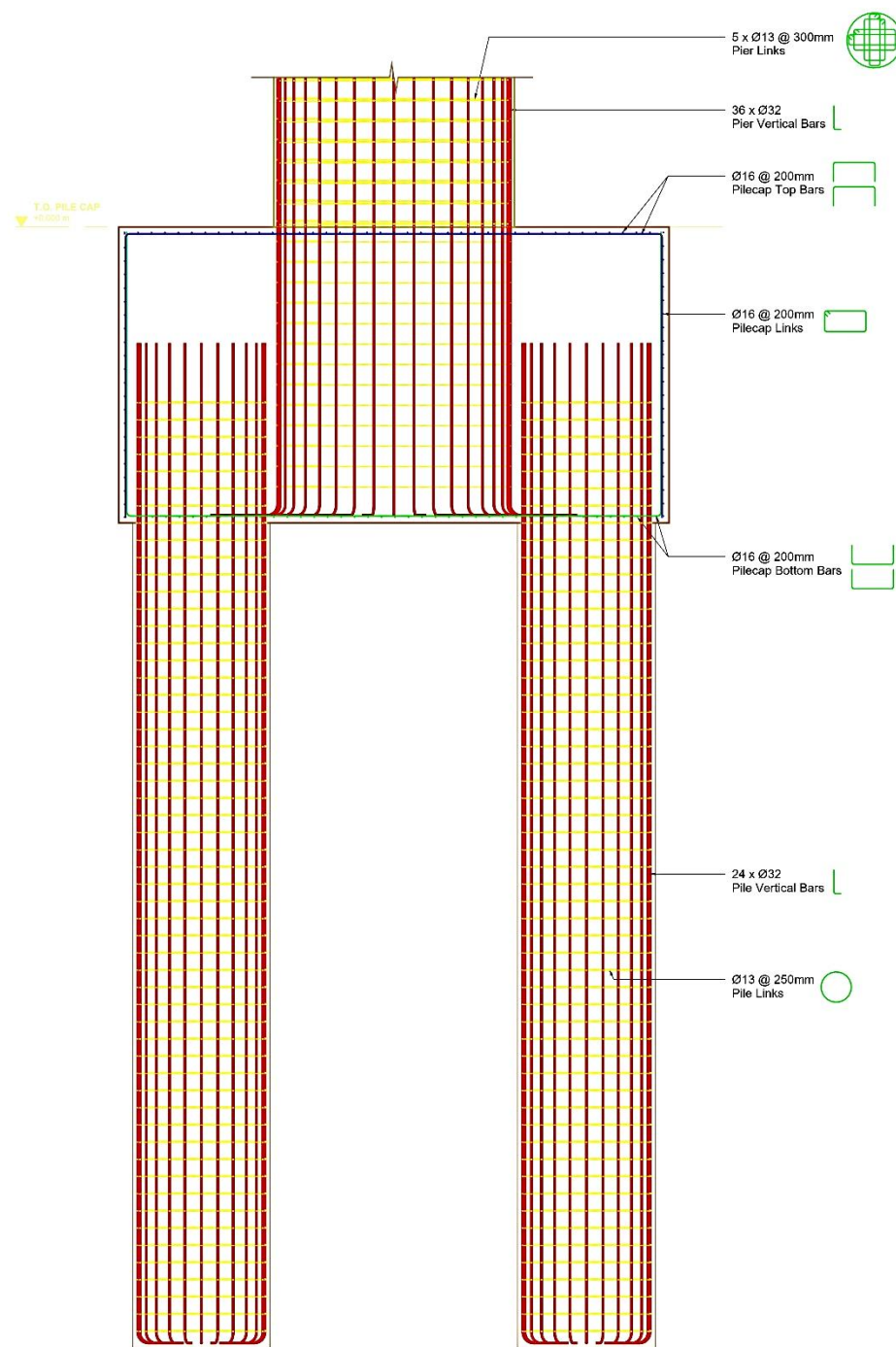
1 REINFORCEMENT PLAN
1 : 50



2 REINFORCEMENT PILE
1 : 50



3 REINFORCEMENT PIER
1 : 50



4 REINFORCEMENT SECTION
1 : 50



- 3D Reinforcement detailing using Revit helps visualize bar arrangement clearly in both pile and pile cap.
- Generated reinforcement plans and sections to ensure accurate bar spacing, sizes, and layout as per design.
- Improved coordination and documentation by integrating structural detailing directly within the BIM environment.

SOFTWARE USED



Autodesk
Revit 2024

SCHEDULES

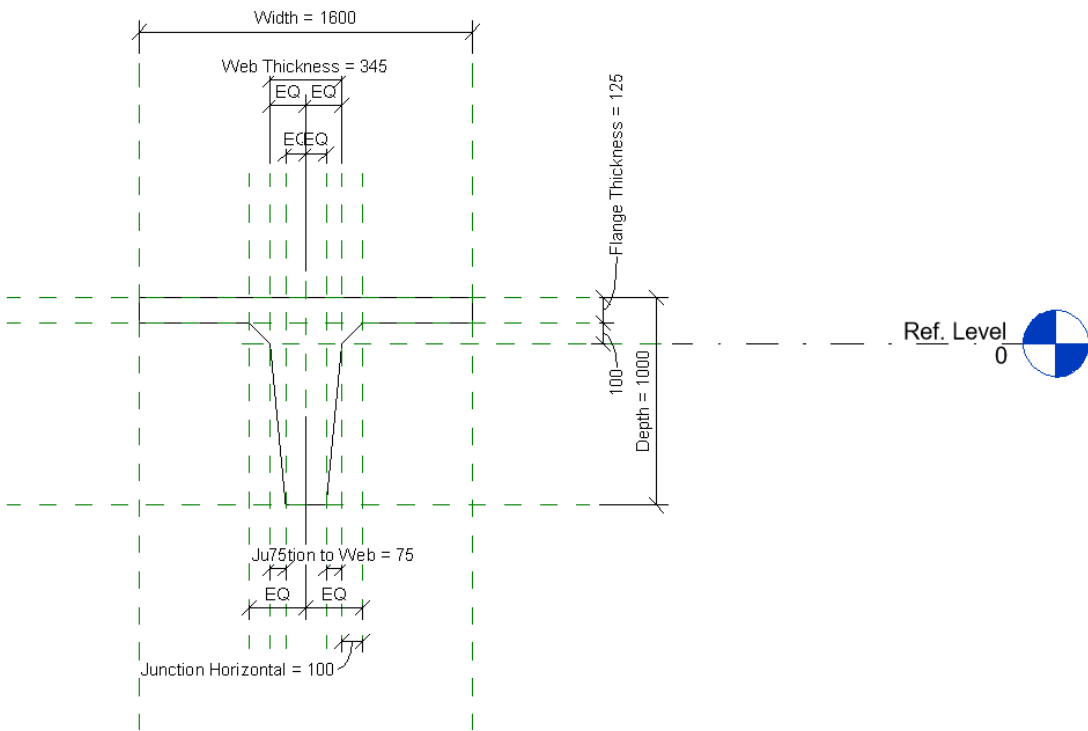
Floor Schedule					
Type	Family	Level	Default Thickness	Area	Count
STU-GEN-SLAB-THK150-CONC	Floor	GROUND FLOOR	150	6 m²	1
STU-GEN-SLAB-THK150-CONC	Floor	GROUND FLOOR	150	9 m²	1
STU-GEN-SLAB-THK150-CONC	Floor	GROUND FLOOR	150	7 m²	1
STU-GEN-SLAB-THK150-CONC	Floor	GROUND FLOOR	150	6 m²	1
STU-GEN-SLAB-THK150-CONC	Floor	GROUND FLOOR	150	9 m²	1
STU-GEN-SLAB-THK150-CONC	Floor	GROUND FLOOR	150	6 m²	1
STU-GEN-SLAB-THK200-CONC	Floor	GROUND FLOOR	200	58 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	23 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	5 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	7 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	15 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	48 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	80 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	6 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	4 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	32 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	63 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	22 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	155 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	12 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	4 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	12 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	4 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	12 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	67 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	11 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	4 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	11 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	4 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	11 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	71 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	47 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	30 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	22 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	GROUND FLOOR	250	36 m²	1
STU-GEN-SLAB-THK300-CONC	Floor	GROUND FLOOR	300	26 m²	1
GROUND FLOOR					36
J056_Concrete Slab 100mm	Floor	CONCOURSE LEVEL	100	392 m²	1
J056_Concrete Slab 100mm	Floor	CONCOURSE LEVEL	100	369 m²	1
J056_Concrete Slab 100mm	Floor	CONCOURSE LEVEL	100	315 m²	1
J056_Concrete Slab 100mm	Floor	CONCOURSE LEVEL	100	335 m²	1
J056_Concrete Slab 100mm	Floor	CONCOURSE LEVEL	100	369 m²	1
J056_Concrete Slab 100mm	Floor	CONCOURSE LEVEL	100	368 m²	1
J056_Concrete Slab 100mm	Floor	CONCOURSE LEVEL	100	368 m²	1
J056_Concrete Slab 100mm	Floor	CONCOURSE LEVEL	100	368 m²	1
J056_Concrete Slab 100mm	Floor	CONCOURSE LEVEL	100	368 m²	1
J056_Concrete Slab 100mm	Floor	CONCOURSE LEVEL	100	110 m²	1
J056_Concrete Slab 100mm	Floor	CONCOURSE LEVEL	100	27 m²	1
STU-GEN-SLAB-THK200-CONC	Floor	CONCOURSE LEVEL	200	15 m²	1
STU-GEN-SLAB-THK200-CONC	Floor	CONCOURSE LEVEL	200	15 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	CONCOURSE LEVEL	250	11 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	CONCOURSE LEVEL	250	209 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	CONCOURSE LEVEL	250	8 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	CONCOURSE LEVEL	250	195 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	CONCOURSE LEVEL	250	12 m²	1
STU-GEN-SLAB-THK300-CONC	Floor	CONCOURSE LEVEL	300	117 m²	1
STU-GEN-SLAB-THK300-CONC	Floor	CONCOURSE LEVEL	300	32 m²	1
STU-GEN-SLAB-THK300-CONC	Floor	CONCOURSE LEVEL	300	36 m²	1
STU-GEN-SLAB-THK300-CONC	Floor	CONCOURSE LEVEL	300	7 m²	1
STU-GEN-SLAB-THK300-CONC	Floor	CONCOURSE LEVEL	300	7 m²	1
STU-GEN-SLAB-THK300-CONC	Floor	CONCOURSE LEVEL	300	281 m²	1
STU-GEN-SLAB-THK300-CONC	Floor	CONCOURSE LEVEL	300	71 m²	1
STU-GEN-SLAB-THK300-CONC	Floor	CONCOURSE LEVEL	300	18 m²	1
CONCOURSE LEVEL					25
STU-GEN-SLAB-THK50-CONC	Floor	UNDER PLATFORM LEVEL	50	553 m²	1
STU-GEN-SLAB-THK50-CONC	Floor	UNDER PLATFORM LEVEL	50	553 m²	1
STU-GEN-SLAB-THK100-CONC	Floor	UNDER PLATFORM LEVEL	100	128 m²	1
STU-GEN-SLAB-THK100-CONC	Floor	UNDER PLATFORM LEVEL	100	226 m²	1
STU-GEN-SLAB-THK100-CONC	Floor	UNDER PLATFORM LEVEL	100	129 m²	1
STU-GEN-SLAB-THK100-CONC	Floor	UNDER PLATFORM LEVEL	100	121 m²	1
STU-GEN-SLAB-THK100-CONC	Floor	UNDER PLATFORM LEVEL	100	121 m²	1
STU-GEN-SLAB-THK100-CONC	Floor	UNDER PLATFORM LEVEL	100	94 m²	1
STU-GEN-SLAB-THK100-CONC	Floor	UNDER PLATFORM LEVEL	100	128 m²	1
STU-GEN-SLAB-THK100-CONC	Floor	UNDER PLATFORM LEVEL	100	226 m²	1
STU-GEN-SLAB-THK100-CONC	Floor	UNDER PLATFORM LEVEL	100	129 m²	1
STU-GEN-SLAB-THK100-CONC	Floor	UNDER PLATFORM LEVEL	100	121 m²	1
STU-GEN-SLAB-THK100-CONC	Floor	UNDER PLATFORM LEVEL	100	100 m²	1
STU-GEN-SLAB-THK200-CONC	Floor	UNDER PLATFORM LEVEL	200	10 m²	1
STU-GEN-SLAB-THK200-CONC	Floor	UNDER PLATFORM LEVEL	200	10 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	UNDER PLATFORM LEVEL	250	21 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	UNDER PLATFORM LEVEL	250	21 m²	1
UNDER PLATFORM LEVEL					18
STU-GEN-SLAB-THK250-CONC	Floor	PLATFORM LEVEL	250	437 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	PLATFORM LEVEL	250	176 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	PLATFORM LEVEL	250	310 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	PLATFORM LEVEL	250	437 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	PLATFORM LEVEL	250	176 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	PLATFORM LEVEL	250	310 m²	1
PLATFORM LEVEL					6
STU-GEN-SLAB-THK200-CONC	Floor	PLATFORM ROOF LEVEL 200	200	6 m²	1
STU-GEN-SLAB-THK200-CONC	Floor	PLATFORM ROOF LEVEL 200	200	16 m²	1
STU-GEN-SLAB-THK200-CONC	Floor	PLATFORM ROOF LEVEL 200	200	16 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	PLATFORM ROOF LEVEL 250	250	184 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	PLATFORM ROOF LEVEL 250	250	419 m²	1
STU-GEN-SLAB-THK250-CONC	Floor	PLATFORM ROOF LEVEL 250	250	419 m²	1
PLATFORM ROOF LEVEL					6
Grand total					91



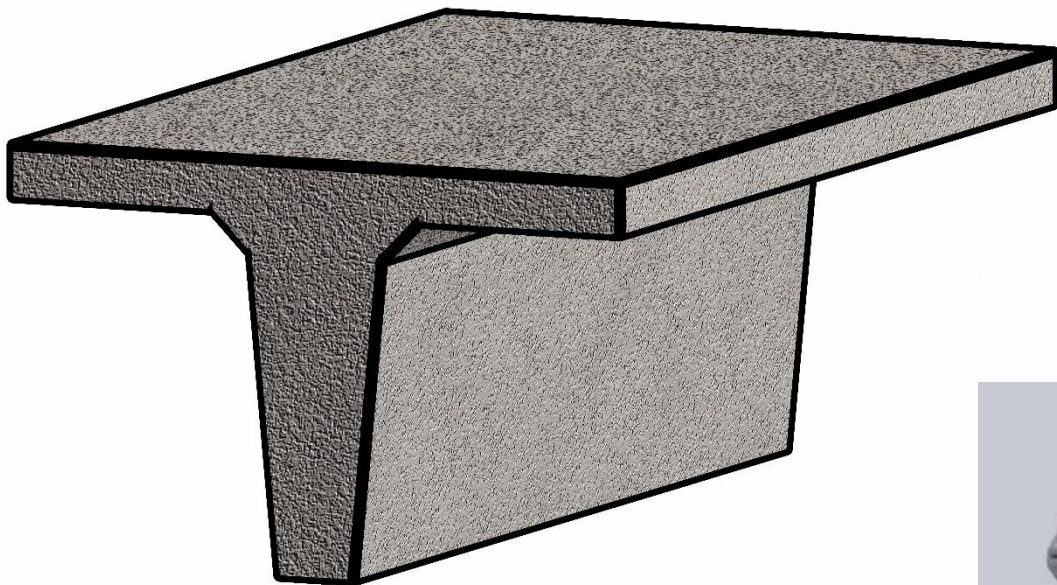
- Learned to create quantitative schedules directly from the model for accurate tracking of materials and elements.
- Explored the use of formulas within schedules to automate cost and quantity calculations.
- Improved understanding of data extraction from BIM models for efficient project planning and documentation.

CREATING CUSTOM FAMILY

TRESTLE BEAM FAMILY

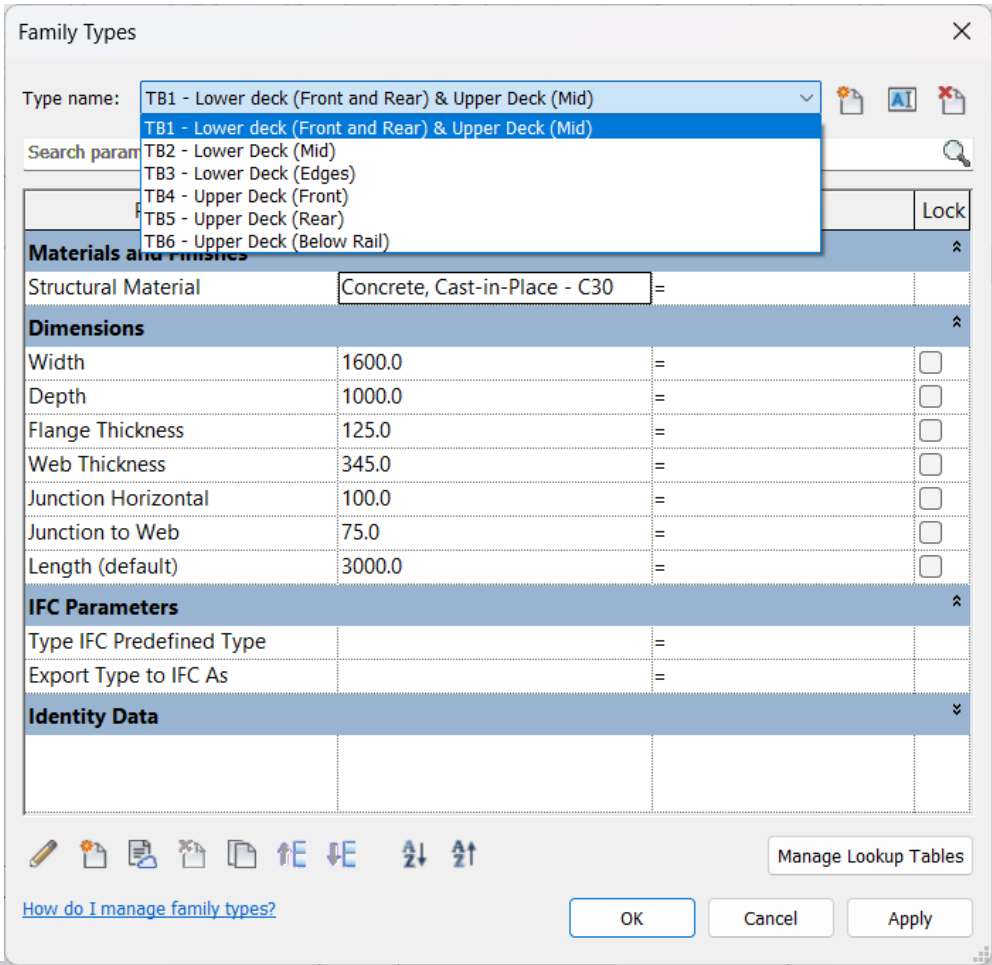


3D VIEW – TRESTLE BEAM



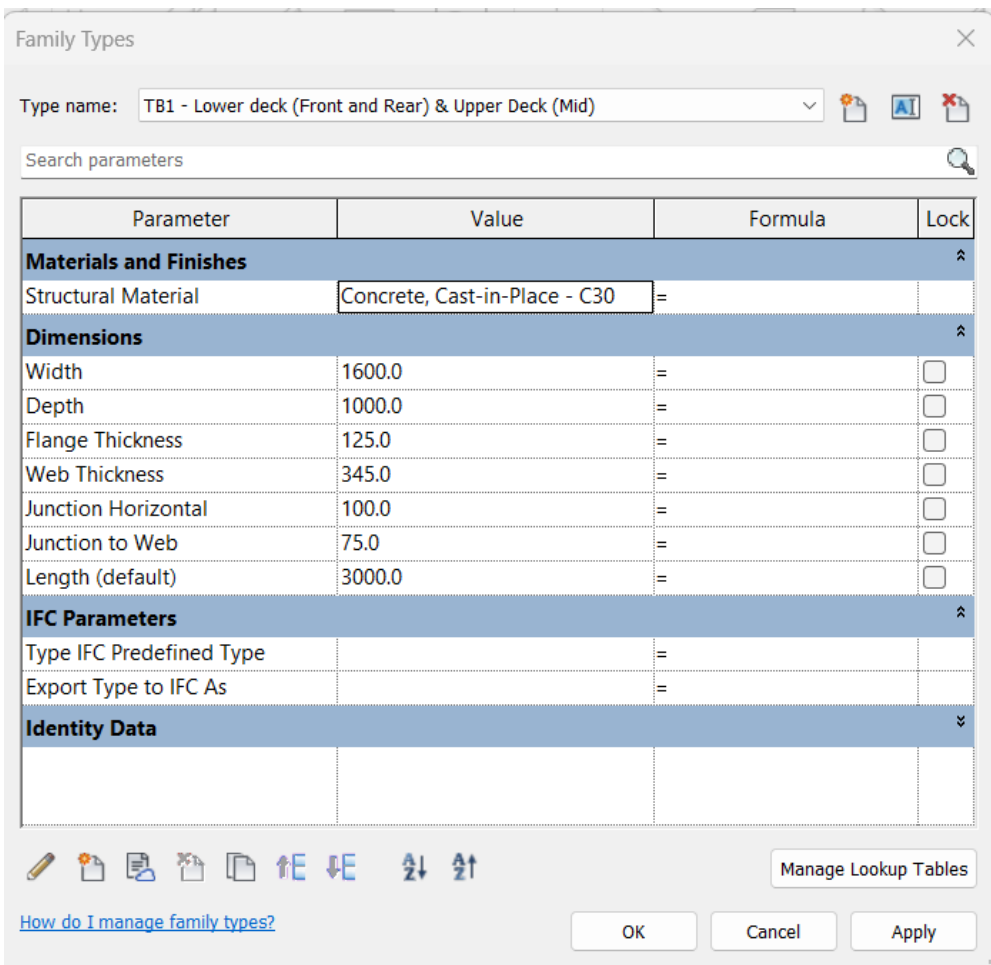
TRESTLE BEAM TYPES

- Created a parametric Trestle Beam family with 6 type variations (TB1 to TB6) based on position and geometry.
- Controlled key dimensions like Width, Depth, Flange Thickness, and Length using type parameters.
- Enabled easy customization and efficient reuse across the BIM model.



PARAMETER CREATED

- Created parameters to control essential family dimensions.
- Enabled flexible adjustments without redrawing the model.
- Improved overall accuracy and efficiency in the BIM workflow.



- Gained hands-on experience in creating custom parametric families with multiple type variations for structural components.
- Improved understanding of parameter control to achieve dimension flexibility and enhance BIM model efficiency.

CDE – TRIMBLE CONNECT

Trimble Connect 3D Viewer - Civil

https://web.connect.trimble.com/projects/ISb1RIPt4/viewer/3d/?viewId=fKIQ00c3-Cw&=&origin=app31.connect.trimble.com

Back

Navigation icons

Reset model

CIVIL-1

Baseplate bolted details

Detailed bolted connection for the steel column is required.

Coordination

Assigned To: gowtham g

Author: gowtham g

Low

New

Request

Jun 30, 2025

Created Jun 19, 2025

Modified Jun 19, 2025 By gowtham g

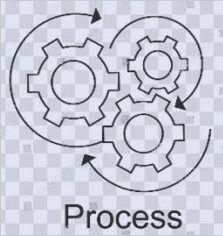
VIEW

Thumbnail

View content for: All objects

GROUP BY: Name

Name	Count	Class	Material	Area	Σ	Volume	Σ	Slope	Σ	Slope
> B3-2:B3-2:1820195 (1)	1	IFCBUILDINGELEMENTPR...	<Unnamed>	2.65 m²		0.25 m³		--		--
> B3-3:B3-3:1820786 (1)	1	IFCBUILDINGELEMENTPR...	<Unnamed>	2.67 m²		0.25 m³		--		--
> B3-4:B3-4:1822145 (1)	1	IFCBUILDINGELEMENTPR...	<Unnamed>	2.68 m²		0.26 m³		--		--
> B3-Staircase1:B3-Staircase1:179909...	1	IFCBUILDINGELEMENTPR...	<Unnamed>	2.27 m²		0.21 m³		--		--



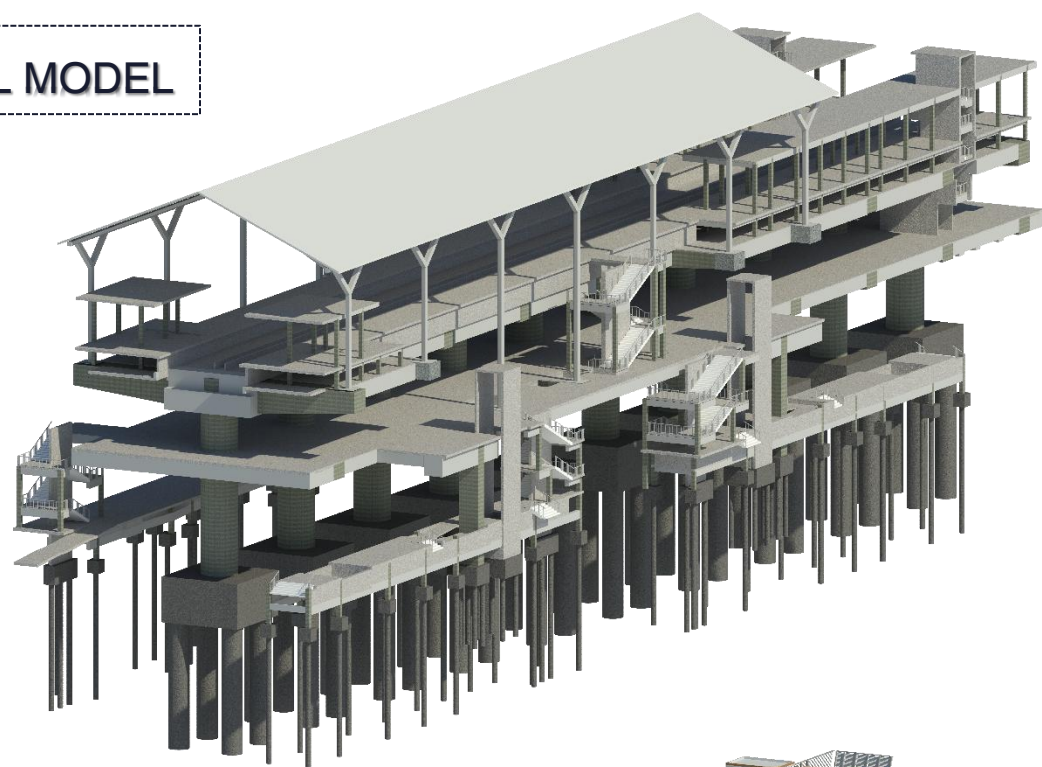
- **Uploaded the federated model** to Trimble Connect for centralized review.
- **Created To-Do with markups** for baseplate bolts and staircase.
- **Assigned and tracked issues** with due dates and priorities.



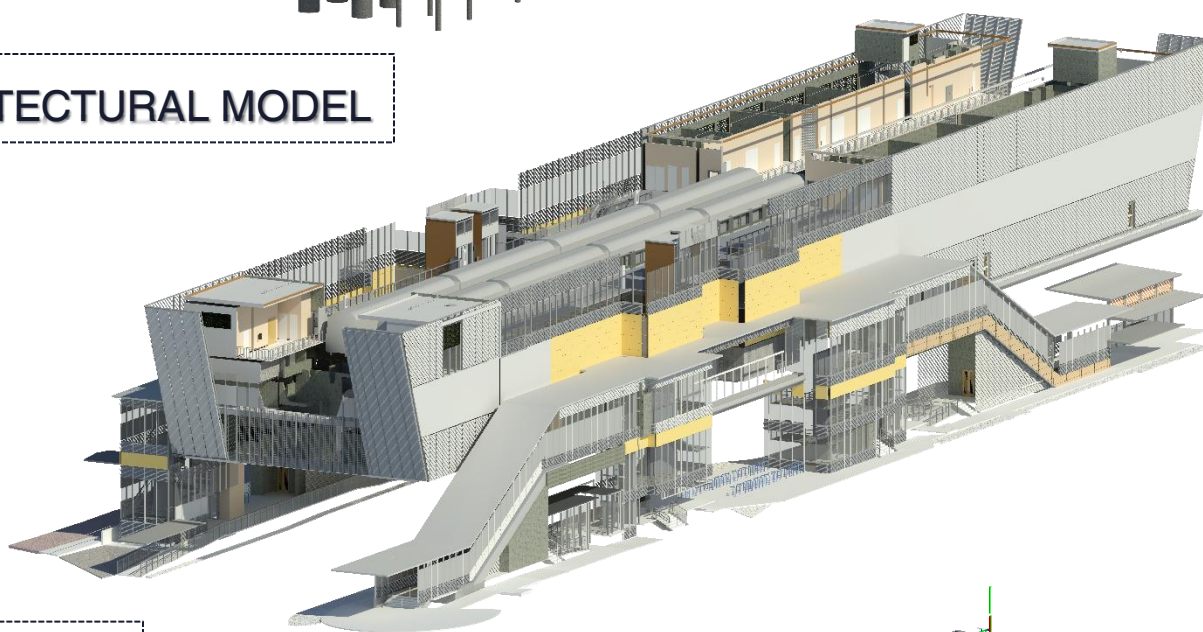
- **Used markup tools** to highlight and communicate design issues.
- Enabled **real-time communication** across the project team.
- Achieved **streamlined workflows** and improved **stakeholder collaboration**.

MODEL CO-ORDINATION

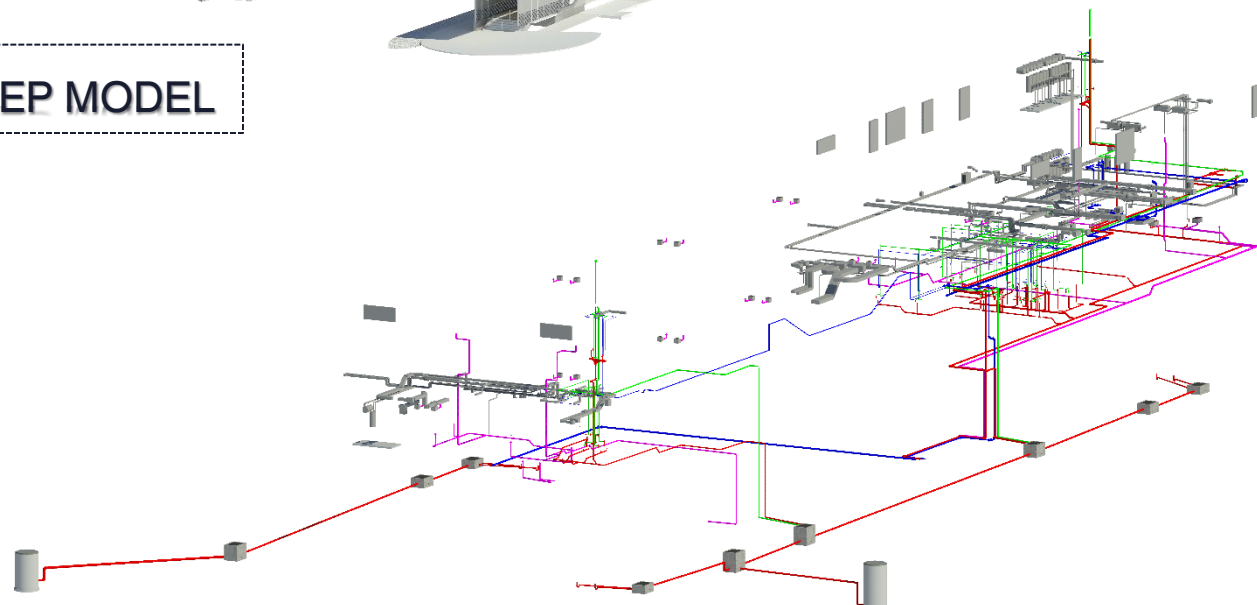
STRUCTURAL MODEL



ARCHITECTURAL MODEL



MEP MODEL



SOFTWARE USED

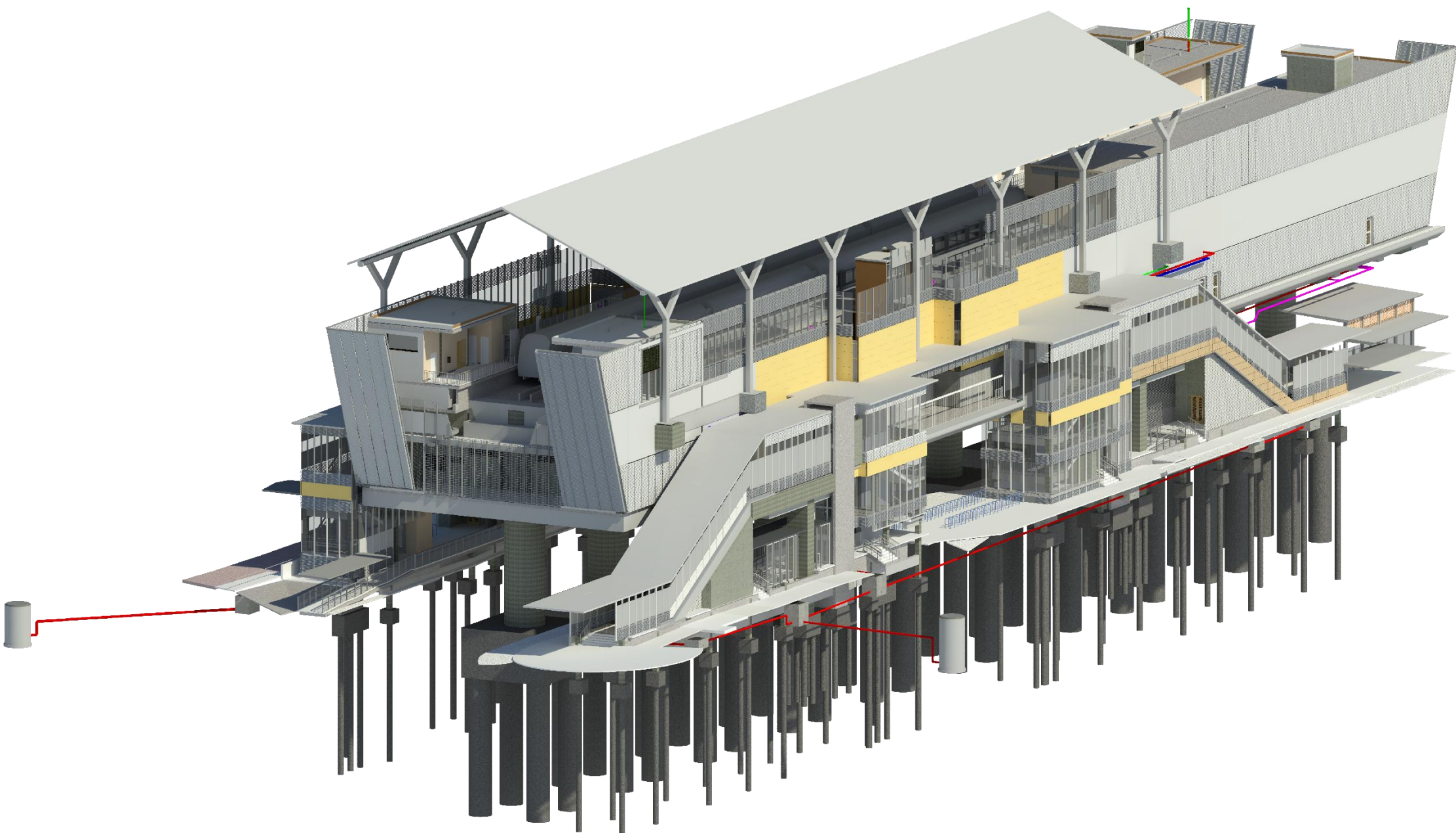


Autodesk
Revit 2024



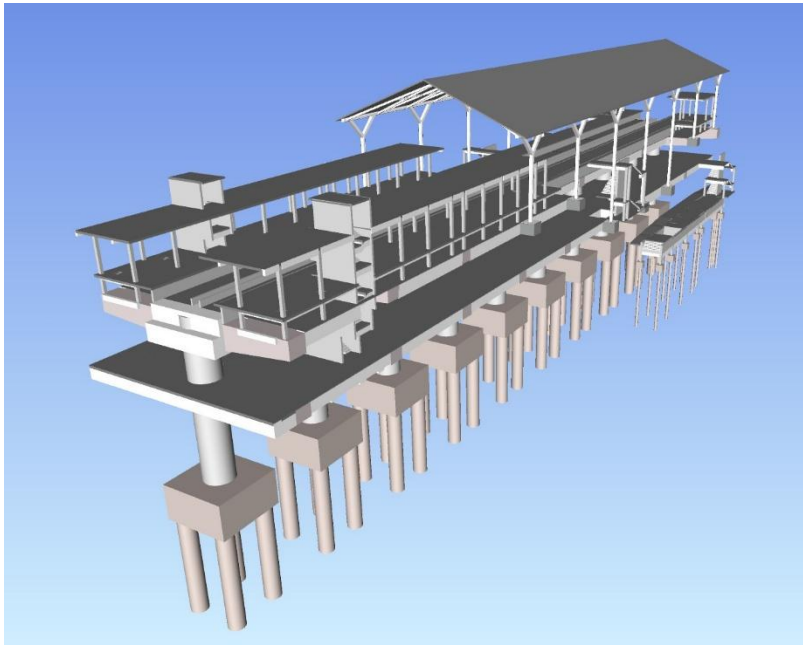
- Understood the process of linking multiple discipline models (Architecture, Structure, MEP) into a single **Federated Model**.
- Learned to manage **file coordination**, resolve **linking issues**, and maintain **model hierarchy**.
- Improved skills in **3D navigation**, **sectioning**, and **visual inspection** to detect overlaps or gaps.

FEDERATED MODEL

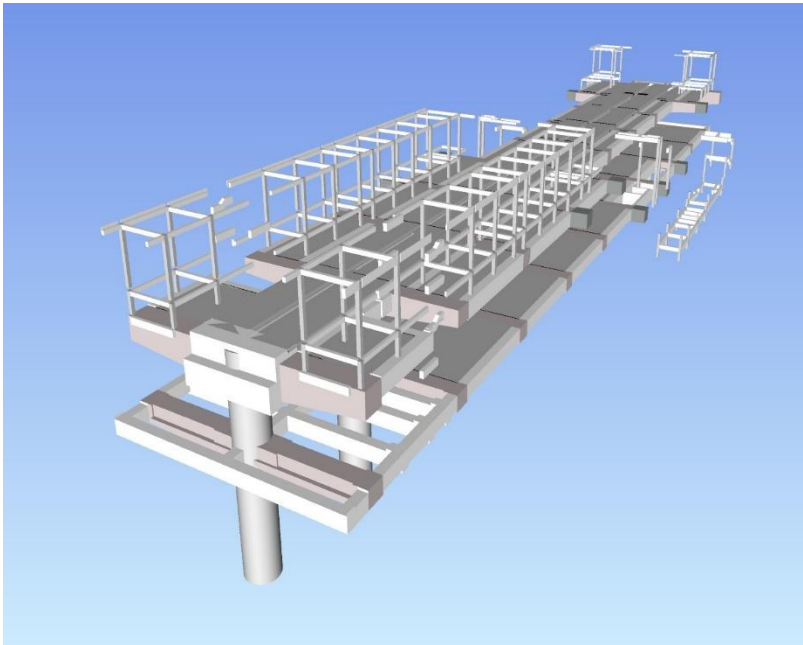


MODEL CO-ORDINATION

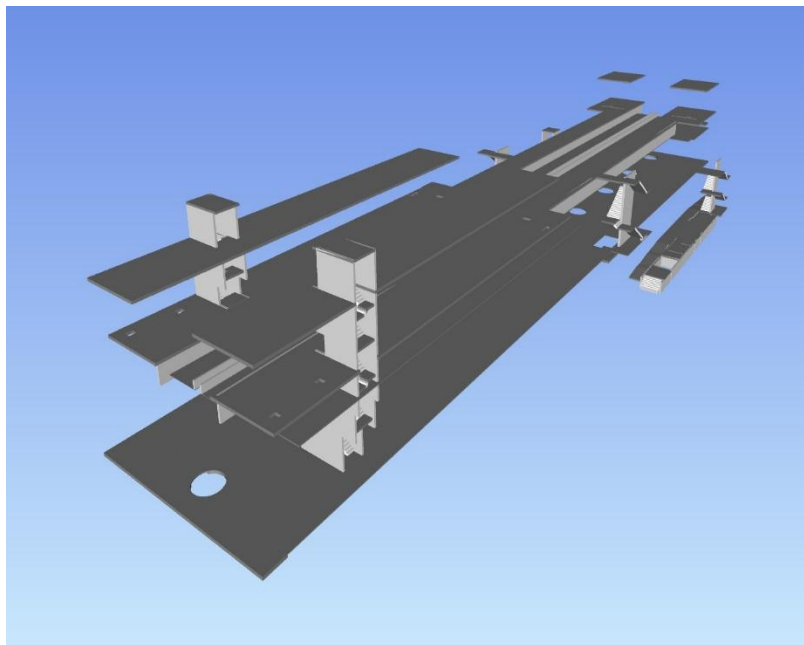
[STR]-Set01



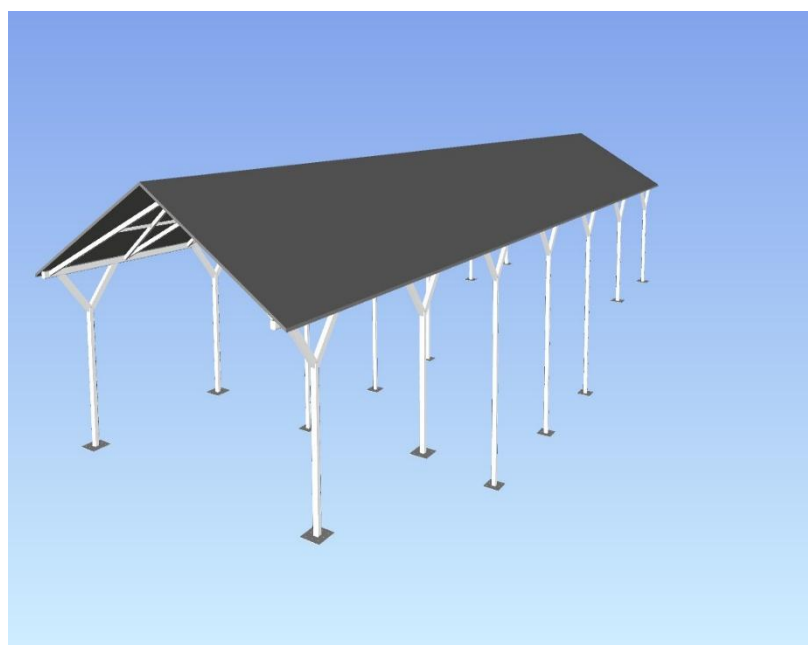
[STR]-Set02



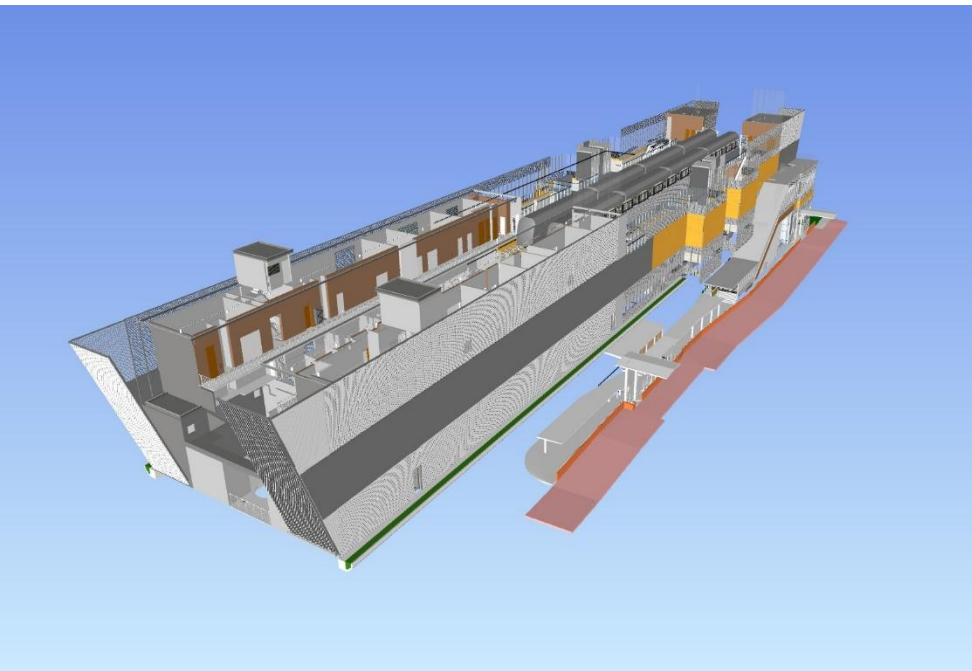
[STR]-Set03



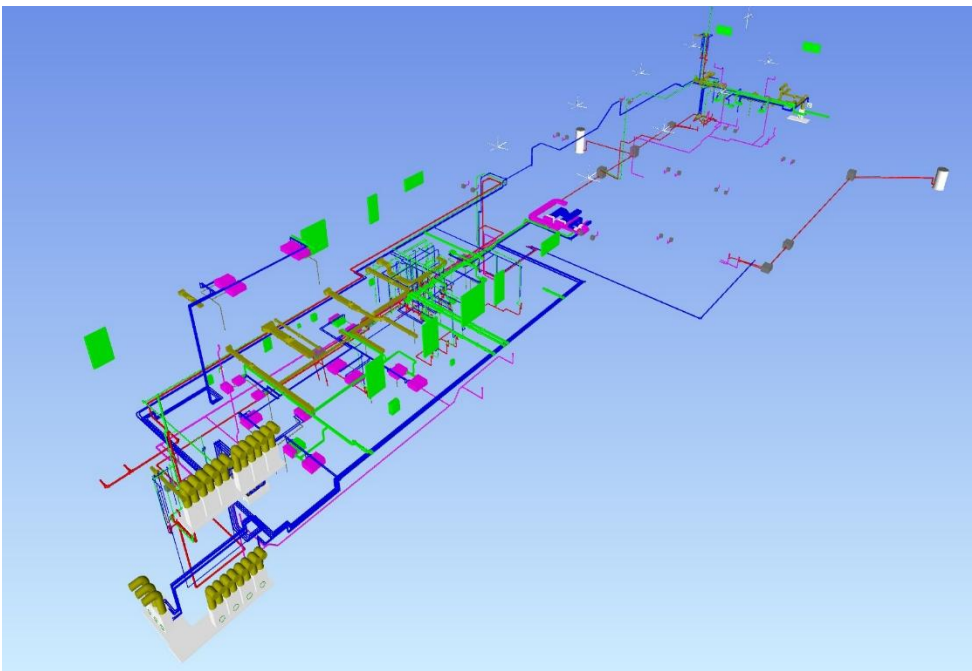
[STR]-Set04



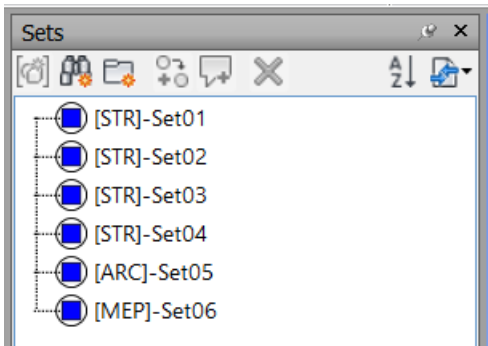
[ARC]-Set05



[MEP]-Set06



DISCIPLINE	NAME OF THE SET	DESCRIPTION
STR-ALL	[STR]-Set01	Structural columns and structural framing (beams) and other elements
STR-Primary Str	[STR]-Set02	Structural columns and structural framing (beams)
STR-Secondary Str	[STR]-Set03	Other (Structural wall, floors, stair, ramps etc.)
STR-Secondary Str	[STR]-Set04	Roof members
ARC-All	[ARC]-Set05	All the Arch elements
MEP-All	[MEP]-Set06	MEP elements



SOFTWARE USED



CLASH DETECTION

CLASH TESTS PRIORITY

Test No	Selection A	Selection B	Test Name	Priority	Rules	Select		
						Type	Link	Tol.
T01	[STR]-Set02	[STR]-Set03	T01_[STR]-Set02-[STR]-Set03	High		Hard	None	5 cm
T02	[STR]-Set01	[ARC]-Set05	T01_[STR]-Set01-[STR]-Set05	Low		Hard	None	10 cm
T03	[STR]-Set01	[MEP]-Set06	T01_[STR]-Set01-[MEP]-Set06	High		Hard	None	5 cm
T04	[MEP]-Set06	[MEP]-Set06	T01_[MEP]-Set06-[MEP]-Set06	Low		Hard (Conservative)	None	2 cm

NAVISWORKS CLASH DETECTION MATRIX

CLASH DETECTION MATRIX		STR	STR	STR	STR	ARC	MEP
		STR. ALL	COLUMN & FRAMING	WALLS, FLOORS, STAIRS, RAMP	ROOF MEMBERS	ARCH. ALL	MEP
STR	STR. ALL	NA				T02	T03
STR	COLUMN & FRAMING		NA	T01			
STR	WALLS, FLOORS, STAIRS, RAMP			NA			
STR	ROOF MEMBERS				NA		
ARC	ARCH. ALL					NA	
MEP	MEP						T04
N/A							
Low Priority							
High Priority							

CLASH DETECTED

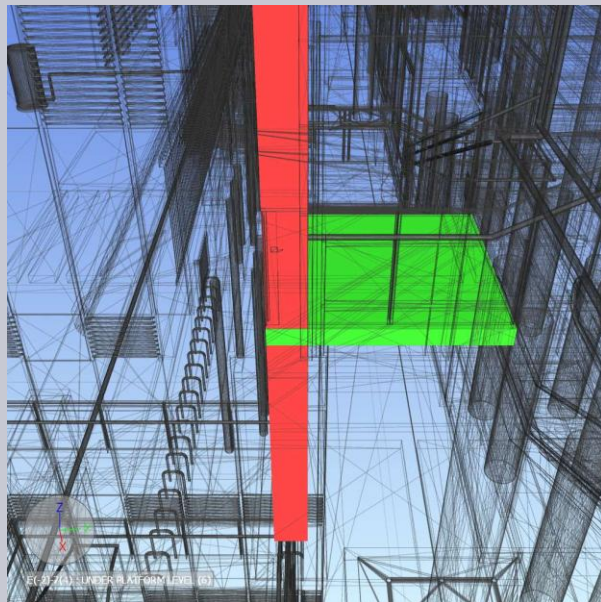
Last Run: 19 June 2025 20:09:47

Clashes - Total: 34 (Open: 34 Closed: 0)

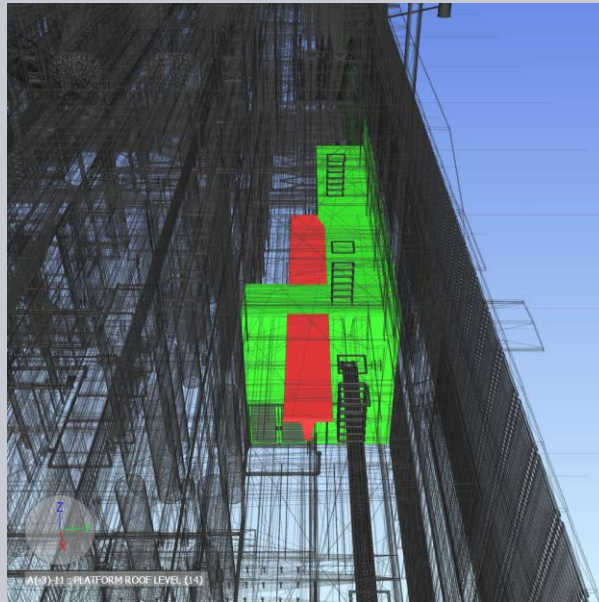
Name	Status	Clashes	New	Active	Reviewed	Approved	Resolved
T01	Done	34	34	0	0	0	0
T02	Done	2340	1170	0	0	0	1170
T03	Done	279	279	0	0	0	0
T04	Done	1317	1317	0	0	0	0

OUTPUT

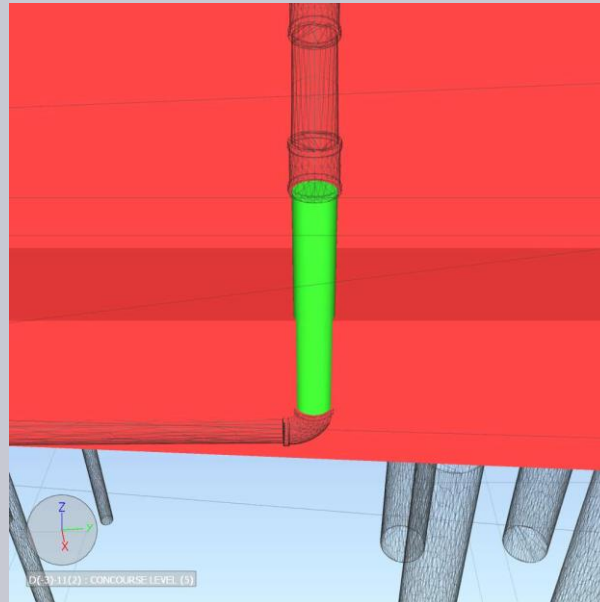
T01_[STR]-Set02-[STR]-Set03



T01_[STR]-Set01-[STR]-Set05



T01_[STR]-Set01-[STR]-Set06



AUTODESK[®]
NAVISWORKS[®]

Clash Report

CLASH DETECTION REPORT

T01	Tolerance	Clashes	New	Active	Reviewed	Approved	Resolved	Type	Status
	0.050m	34	34	0	0	0	0	Hard	OK

Image	Clash Name	Status	Distance	Grid Location	Description	Date Found	Clash Point	Item 1	Item 2
								Item ID	Item ID
	Clash1	New	-0.401	F-7 : CONCOURSE LEVEL	Hard	2025/6/19 14:39	x:12941.912, y:36704.949, z:14.690	Element ID: 1704805	GUID: a5acefed-d8f2-4fff-be32-7246a8ecb35a
	Clash2	New	-0.400	A-7 : CONCOURSE LEVEL	Hard	2025/6/19 14:39	x:12946.502, y:36735.383, z:14.890	Element ID: 1705287	GUID: 53399492-6b67-4db4-bc43-1765331f6fc6
	Clash3	New	-0.387	E-7 : CONCOURSE LEVEL	Hard	2025/6/19 14:39	x:12943.060, y:36705.067, z:14.890	Element ID: 1704643	GUID: a5acefed-d8f2-4fff-be32-7246a8ecb35a
	Clash4	New	-0.320	F-7 : GROUND FLOOR	Hard	2025/6/19 14:39	x:12938.947, y:36699.760, z:5.440	Element ID: 1756761	GUID: 13a04806-9649-4368-84dc-c7c845c8076a

- Clash Detection Matrix was used to create specific task sets based on project requirements and tolerance limits.
- Clashes within disciplines were identified and categorized according to the defined task sets.
- Issues were resolved in the Revit model using the Switchback feature for seamless coordination.

SOFTWARE USED



CLASH RESOLUTION

- After generating the **Clash Report**, identified issues were categorized by discipline.
- **Architecture and MEP clashes** were **assigned** to their respective teams for resolution.
- All **Structural clashes** were successfully **resolved** within the Revit model.

⬆️ T01

CLASH RESOLUTION REPORT

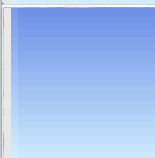
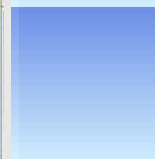
Last Run: 19 June 2025 20:09:47

Clashes - Total: 34 (Open: 0 Closed: 34)

	Name	Status	Clashes	New	Active	Reviewed	Approved	Resolved	
	T01	Done	0	0	0	0	0	34	
	T02	Done	2340	0	2339	0	1	0	
	T03	Done	279	0	279	0	0	0	
	T04	Done	1317	0	1317	0	0	0	

AUTODESK® NAVISWORKS® Clash Report

T01	Tolerance	Clashes	New	Active	Reviewed	Approved	Resolved	Type	Status
	0.050m	0	0	0	0	0	34	Hard	OK

									Item 1	Item 2
Image	Clash Name	Status	Distance	Grid Location	Description	Date Found	Assigned To	Clash Point	Item ID	Item ID
	Clash1	Resolved	-0.401	F-7 : CONCOURSE LEVEL	Hard	2025/6/19 14:39		x:12941.912, y:36704.949, z:14.690	Element ID: 1704805	GUID: a5acefed-d8f2-4fff-be32-7246a8ecb35a
	Clash2	Resolved	-0.400	A-7 : CONCOURSE LEVEL	Hard	2025/6/19 14:39		x:12946.502, y:36735.383, z:14.890	Element ID: 1705287	GUID: 53399492-6b67-4db4-bc43-1765331f6fc6
	Clash3	Resolved	-0.387	E-7 : CONCOURSE LEVEL	Hard	2025/6/19 14:39		x:12943.060, y:36705.067, z:14.890	Element ID: 1704643	GUID: a5acefed-d8f2-4fff-be32-7246a8ecb35a
	Clash4	Resolved	-0.320	F-7 : GROUND FLOOR	Hard	2025/6/19 14:39		x:12938.947, y:36699.760, z:5.440	Element ID: 1756761	GUID: 13a04806-9649-4368-84dc-c7c845c8076a

4D SIMULATION

4D PROJECT PARAMETER

Created a 4D Project Parameter as an instance-based text parameter in Revit, and assigned Activity IDs to each element for accurate 4D and phase-wise planning.

Parameter Properties

Parameter Type

☒ Project parameter

(Can appear in schedules but not in tags)

☐ Shared parameter

(Can be shared by multiple projects and families, exported to ODBC, and appear in schedules and tags)

Select...

Export...

Parameter Data

Name:

4D

Discipline:

Common

Data Type:

Text

Group parameter under:

Identity Data

Tooltip Description:

<No tooltip description. Edit this parameter to write a custom tooltip. Custom tooltips have a limit of 250 characters.

Edit Tooltip...

Categories

Category name search:

Filter list:

Structure

☐ Hide un-checked categories

☐ Analytical Links

☐ Analytical Members

☐ Analytical Nodes

☐ Analytical Openings

☐ Analytical Panels

☐ Assemblies

☐ Audio Visual Devices

☐ Columns

☐ Data Exchanges

☐ Detail Items

☐ Fire Protection

☒ Floors

☐ Food Service Equipment

☒ Generic Models

☐ Grids

☐ Levels

☒ Lines

Check All

Check None

☒ Add to all elements in the selected categories

OK

Cancel

Help

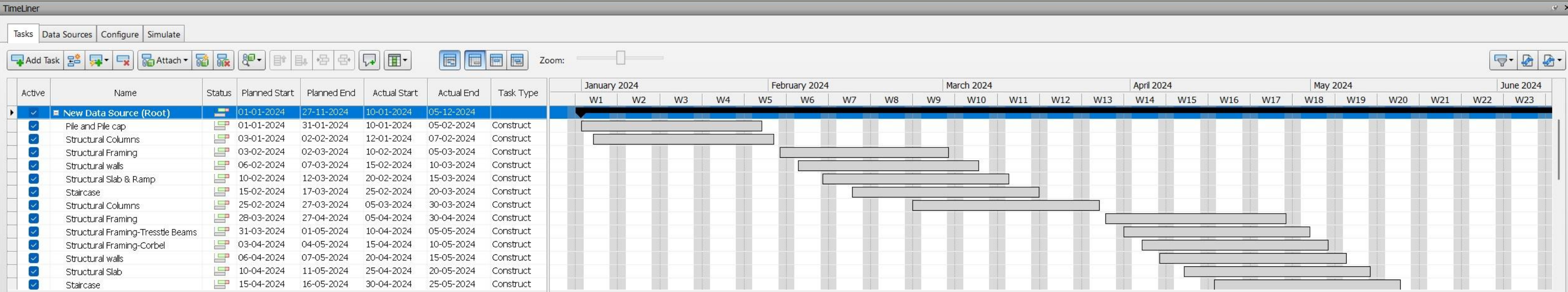
CONSTRUCTION SCHEDULE

The following represents the work schedule given data

Activity ID	Activity Status	WBS Code	(*)WBS Name	Activity (Task) Name	Original Duration(d)	(*)Actual Duration(d)	(*)Planned Start	(*)Planned End	(*)Actual Start	(*)Actual End	Task Type
00.00.001	Started	00-00	Foundation	Pile and Pile cap	30	26	01-01-2024	31-01-2024	10-01-2024	05-02-2024	Construct
00.01.001	Started	00-01	Ground floor Level	Structural Columns	30	26	03-01-2024	02-02-2024	12-01-2024	07-02-2024	Construct
00.01.002	Started	00-01	Ground floor Level	Structural Framing	28	24	03-02-2024	02-03-2024	10-02-2024	05-03-2024	Construct
00.01.003	Started	00-01	Ground floor Level	Structural walls	30	24	06-02-2024	07-03-2024	15-02-2024	10-03-2024	Construct
00.01.004	Started	00-01	Ground floor Level	Structural Slab & Ramp	31	24	10-02-2024	12-03-2024	20-02-2024	15-03-2024	Construct
00.01.005	Started	00-01	Ground floor Level	Staircase	31	24	15-02-2024	17-03-2024	25-02-2024	20-03-2024	Construct
00.02.001	Started	00-02	Concourse	Structural Columns	31	25	25-02-2024	27-03-2024	05-03-2024	30-03-2024	Construct
00.02.002	Started	00-02	Concourse	Structural Framing	30	25	28-03-2024	27-04-2024	05-04-2024	30-04-2024	Construct
00.02.003	Started	00-02	Concourse	Structural Framing-Tresstle Beams	31	25	31-03-2024	01-05-2024	10-04-2024	05-05-2024	Construct
00.02.004	Started	00-02	Concourse	Structural Framing-Corbel	31	25	03-04-2024	04-05-2024	15-04-2024	10-05-2024	Construct
00.02.005	Started	00-02	Concourse	Structural walls	31	25	06-04-2024	07-05-2024	20-04-2024	15-05-2024	Construct
00.02.006	Started	00-02	Concourse	Structural Slab	31	25	10-04-2024	11-05-2024	25-04-2024	20-05-2024	Construct
00.02.007	Started	00-02	Concourse	Staircase	31	25	15-04-2024	16-05-2024	30-04-2024	25-05-2024	Construct
00.03.001	Started	00-03	Under Platform level	Structural Columns	1	15	20-06-2024	21-06-2024	30-06-2024	15-07-2024	Construct
00.03.002	Started	00-03	Under Platform level	Structural Framing	32	25	28-05-2024	29-06-2024	05-06-2024	30-06-2024	Construct
00.03.003	Started	00-03	Under Platform level	Structural Framing-Tresstle Beams	31	25	01-06-2024	02-07-2024	10-06-2024	05-07-2024	Construct
00.03.004	Started	00-03	Under Platform level	Structural Framing-Corbel	31	25	03-06-2024	04-07-2024	15-06-2024	10-07-2024	Construct
00.03.005	Started	00-03	Under Platform level	Structural walls	31	25	06-06-2024	07-07-2024	20-06-2024	15-07-2024	Construct
00.03.006	Started	00-03	Under Platform level	Structural Slab	31	25	10-06-2024	11-07-2024	25-06-2024	20-07-2024	Construct
00.03.007	Started	00-03	Under Platform level	Staircase	31	25	15-06-2024	16-07-2024	30-06-2024	25-07-2024	Construct
00.04.001	Started	00-04	Platform level	Structural Columns	31	25	20-06-2024	21-07-2024	05-07-2024	30-07-2024	Construct
00.04.002	Not Started	00-04	Platform level	Structural Framing	32	25	28-07-2024	29-08-2024	05-08-2024	30-08-2024	Construct
00.04.003	Not Started	00-04	Platform level	Structural walls	32	26	01-08-2024	02-09-2024	10-08-2024	05-09-2024	Construct
00.04.004	Not Started	00-04	Platform level	Structural Slab	32	26	03-08-2024	04-09-2024	15-08-2024	10-09-2024	Construct
00.04.005	Not Started	00-04	Platform level	Staircase	32	26	05-08-2024	06-09-2024	20-08-2024	15-09-2024	Construct
00.05.001	Not Started	00-05	Platform Roof	Structural Columns	32	26	10-08-2024	11-09-2024	25-08-2024	20-09-2024	Construct
00.05.002	Not Started	00-05	Platform Roof	Structural Framing	32	25	17-09-2024	19-10-2024	25-09-2024	20-10-2024	Construct
00.05.003	Not Started	00-05	Platform Roof	Structural walls	31	25	21-09-2024	22-10-2024	30-09-2024	25-10-2024	Construct
00.05.004	Not Started	00-05	Platform Roof	Structural Slab	31	25	23-09-2024	24-10-2024	05-10-2024	30-10-2024	Construct
00.06.001	Not Started	00-06	Station Track	Track tresstle beam	31	25	25-09-2024	26-10-2024	10-10-2024	04-11-2024	Construct
00.06.002	Not Started	00-06	Station Track	Track Slab	31	25	08-10-2024	08-11-2024	25-10-2024	19-11-2024	Construct
00.07.001	Not Started	00-07	Roof Level	Roof Columns	32	26	12-10-2024	13-11-2024	30-10-2024	25-11-2024	Construct
00.07.002	Not Started	00-07	Roof Level	Roof Framing	42	26	14-10-2024	25-11-2024	04-11-2024	30-11-2024	Construct
00.07.003	Not Started	00-07	Roof Level	Sheet Canopy	42	26	16-10-2024	27-11-2024	09-11-2024	05-12-2024	Construct

TIMELINER

- The given **construction schedule** in **Excel** format was converted to a **CSV** file.
- It was then **imported** into the **TimeLiner** under the **Data Sources** tab in Navisworks.
- The **linked construction schedule** is **displayed below** for reference.

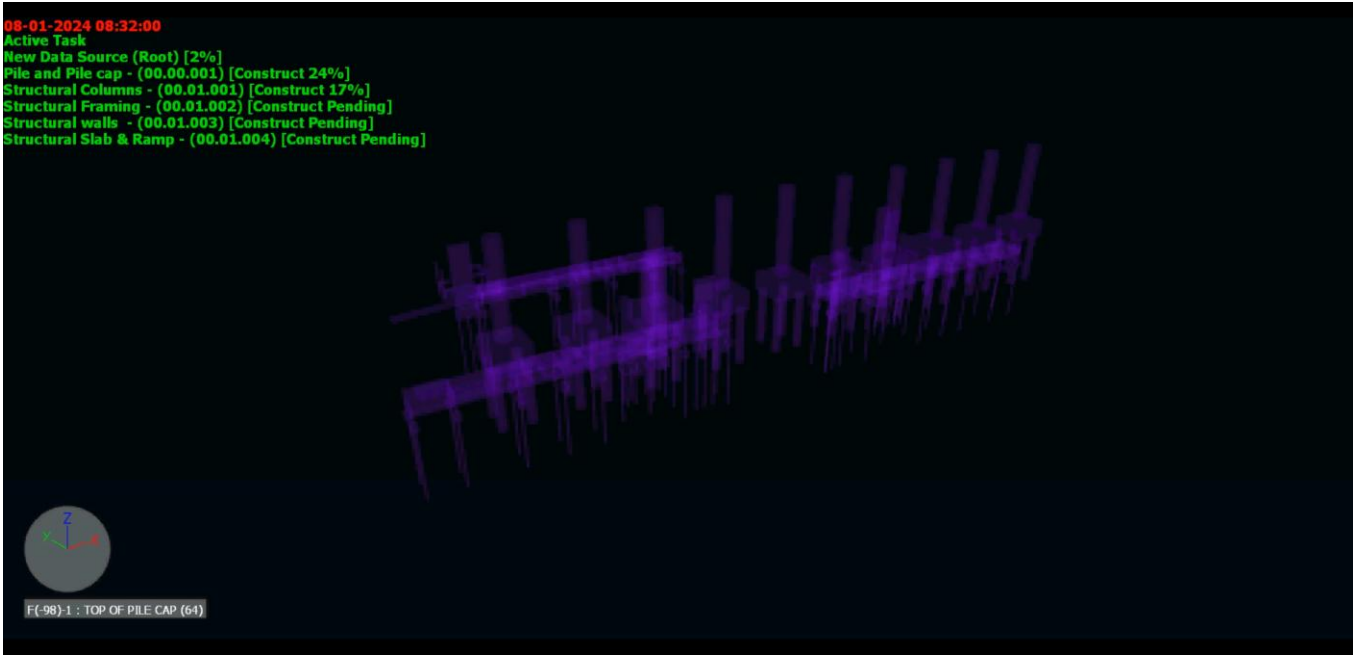


PLANNED VS ACTUAL STIMULATION

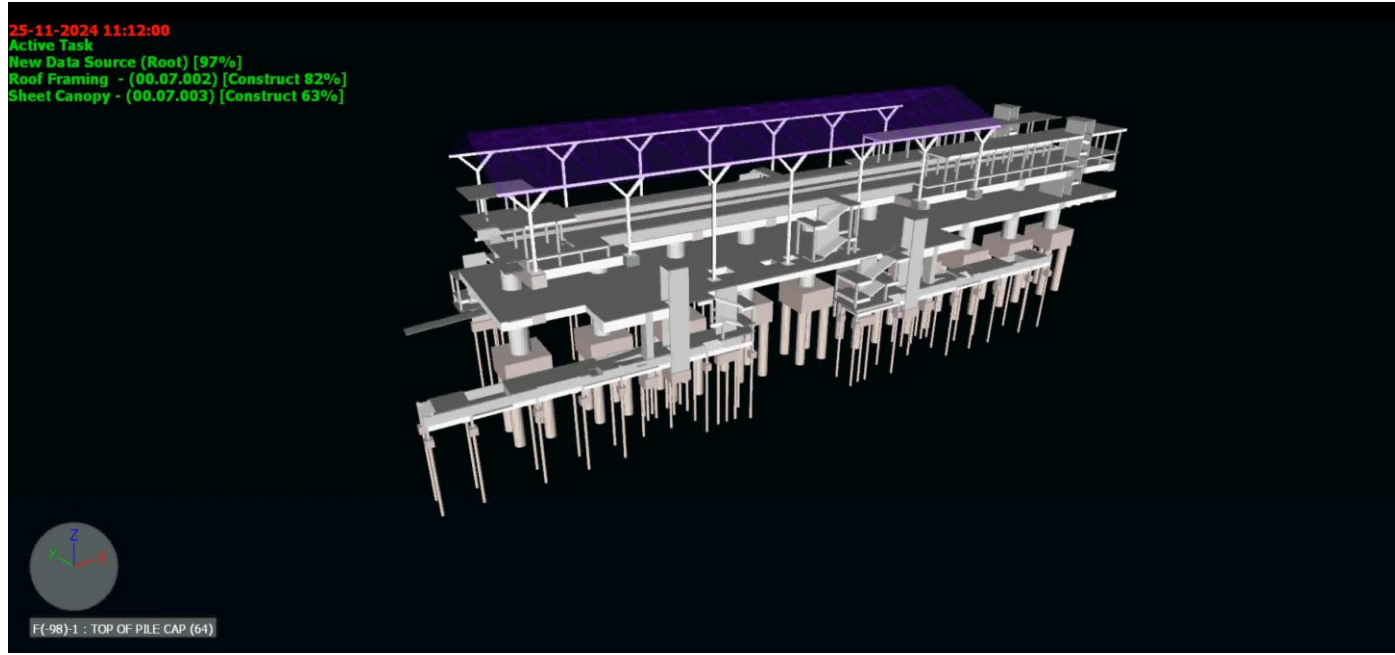
SOFTWARE USED



PLANNED STIMULATION



ACTUAL STIMULATION



PLANNED VS ACTUAL COMPARITIVE STIMULATION

CLICK HERE 

5D QUANTIFICATION

Created a 4D Project Parameter as an instance-based text parameter in Revit, and assigned Activity IDs to each element for accurate quantification and phase-wise planning.

The WBS codes from the provided work schedule are displayed below.

Quantification Workbook												
Select Markup												
Items												
Ground floor Level	00-01	Status	WBS/RBS	Name	Description	Comments	Length	Width	Thickness	Height	Perimeter	Ar
Structural Columns(67)	00-01.01		00-01.01	Structural Columns			423.065 m	0.000 m	0.000 m	0.000 m	0.000 m	
Structural Framing(80)	00-01.02											
Structural walls(96)	00-01.03											
Structural Slab(36)	00-01.04											
Staircase(8)	00-01.05											
Concourse	00-02											
Structural Columns(2)	00-02.01											
Structural Framing(121)	00-02.02											
Structural Framing-Tresstle Beams(89)	00-02.03											
Structural Framing-Corbel(6)	00-02.04											
Structural walls(4)	00-02.05											
Structural Slab(24)	00-02.06											
Staircase(1)	00-02.07											
Under Platform level	00-03											
Platform level	00-04											
Platform Roof	00-05											
Station Track	00-06											
Roof Level	00-07											
Project(0)	00-00											
Filter Markup												
Status												
WBS												
Object												
Viewpoint												
Comments												
ModelLength												
ModelWidth												
ModelThickness												
ModelHeight												
ModelPerimeter												
ModelArea												
00-01.01.1	Concrete Square						3.740 m					
00-01.01.2	Concrete Square						3.740 m					
00-01.01.3	Concrete Square						3.740 m					
00-01.01.4	Concrete Square						3.740 m					
00-01.01.5	Concrete Square						3.740 m					
00-01.01.6	Concrete Square						3.740 m					
00-01.01.7	Concrete Square						3.740 m					
00-01.01.8	Concrete Square						3.740 m					
00-01.01.9	Concrete Square						3.740 m					
00-01.01.10	Concrete Square						3.740 m					
00-01.01.11	Concrete Square						3.140 m					
00-01.01.12	Concrete Square						3.740 m					
00-01.01.13	Concrete Square						3.740 m					
00-01.01.14	Concrete Square						3.740 m					
00-01.01.15	Concrete Square						3.740 m					

Activity ID	Activity Status	WBS Code
00.00.001	Started	00-00
00.01.001	Started	00-01
00.01.002	Started	00-01
00.01.003	Started	00-01
00.01.004	Started	00-01
00.01.005	Started	00-01
00.02.001	Started	00-02
00.02.002	Started	00-02
00.02.003	Started	00-02
00.02.004	Started	00-02
00.02.005	Started	00-02
00.02.006	Started	00-02
00.02.007	Started	00-02
00.03.001	Started	00-03
00.03.002	Started	00-03
00.03.003	Started	00-03
00.03.004	Started	00-03
00.03.005	Started	00-03
00.03.006	Started	00-03
00.03.007	Started	00-03
00.04.001	Started	00-04
00.04.002	Not Started	00-04
00.04.003	Not Started	00-04
00.04.004	Not Started	00-04
00.04.005	Not Started	00-04
00.05.001	Not Started	00-05
00.05.002	Not Started	00-05
00.05.003	Not Started	00-05
00.05.004	Not Started	00-05
00.06.001	Not Started	00-06
00.06.002	Not Started	00-06
00.07.001	Not Started	00-07
00.07.002	Not Started	00-07
00.07.003	Not Started	00-07

QUANTITY REPORT OF STR COLUMN EXPORTED TO EXCEL

The material take-off generated here is utilized for cost estimation.

	A	D	E	F	G	J	K	V	W	Z	AA	AL	AM	AP	AQ
1	WBS/RBS	Group1	Item	Resource	Object	ModelLength	ModelWidth	ModelVolume	ModelPerimeter	ModelArea	Length	Volume	Count	Count	Count
2	00-01	Ground floor Level													
3															
4	00-01.01	Ground floor Level	Structural Columns			423.065 m		1,888.528 m³			67.000 ea				
5	00-01.01.1	Ground floor Level	Structural Columns		Concrete Square	3.740 m		0.598 m³	3.740 m	0.598 m³	1.000 ea				
6	00-01.01.2	Ground floor Level	Structural Columns		Concrete Square (2)	3.740 m		0.598 m³	3.740 m	0.598 m³	1.000 ea				
7	00-01.01.3	Ground floor Level	Structural Columns		Concrete Square (3)	3.740 m		0.380 m³	3.740 m	0.380 m³	1.000 ea				
8	00-01.01.4	Ground floor Level	Structural Columns		Concrete Square (4)	3.740 m		0.380 m³	3.740 m	0.380 m³	1.000 ea				
9	00-01.01.5	Ground floor Level	Structural Columns		Concrete Square (5)	3.740 m		0.357 m³	3.740 m	0.357 m³	1.000 ea				
10	00-01.01.6	Ground floor Level	Structural Columns		Concrete Square (6)	3.740 m		0.479 m³	3.740 m	0.479 m³	1.000 ea				
11	00-01.01.7	Ground floor Level	Structural Columns		Concrete Square (7)	3.740 m		0.479 m³	3.740 m	0.479 m³	1.000 ea				
12	00-01.01.8	Ground floor Level	Structural Columns		Concrete Square (8)	3.740 m		0.357 m³	3.740 m	0.357 m³	1.000 ea				
13	00-01.01.9	Ground floor Level	Structural Columns		Concrete Square (9)	3.740 m		0.598 m³	3.740 m	0.598 m³	1.000 ea				

Let's Build Better with BIM!

Actively seeking BIM Coordinator Opportunities



Connect with me to collaborate or discuss BIM projects



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Thank You!